University of Colorado **Boulder**

Course Schedule

	MON	TUES	WED	THU	FRI
Course overview and Intro				08/20 First day of classes	
Module 1: The Sun and the Solar Wind		08/25 Review of E&M and Plasma		08/27 The Sun and Solar Wind	
		09/01 The Solar Wind and Heliosphere		09/03 The Solar Effects	9/4 HW1 due (midnight)
Module 2: The Atmosphere		09/08 The Atmosphere	9/9 HW1 revision due (midnight); MOD 1 QUIZ, open at 8am	09/10 The Atmosphere	
		09/15 The Atmosphere		09/17 The Atmosphere	9/18 HW2 due (midnight)
Module 3: The Ionosphere		09/22 The Ionosphere	9/23 HW2 revision due (midnight); MOD 2 QUIZ, open at 8am	09/24 The Ionosphere	
		9/29 The Ionosphere		10/01 The Ionosphere	10/02 HW3 due (midnight)
Module 4: The Magnetosphere		10/06 The Magnetosphere	10/07 HW3 revision due (midnight); MOD 3 QUIZ, open at 8am	10/08 Reading Day	
		10/13 The Magnetosphere		10/15 The Magnetosphere	
Module 5: The Radiation Environment		10/20 The Magnetosphere		10/22 Radiation	10/23 HW4 due (midnight)
		10/27 Radiation	10/28 HW4 revision due (midnight); MOD 4 QUIZ, open at 8am	10/29 Radiation	
Module 6: Micrometeoroids and Orbital Debris (MMOD)		11/03 Radiation		11/05 MMOD	11/06 HW5 due (midnight)
		11/10 MMOD	11/11 HW5 revision due (midnight); MOD 5 QUIZ, open at 8am	11/12 MMOD	
Module 7: Comparative Environments		11/17 Mars-Jupiter		11/19 Mars-Jupiter	11/20 HW6 due (midnight)
Thanksgiving Break	11/23	11/24	11/25	11/26	11/27
		12/01 Mars-Jupiter	12/02 HW6 revision due (midnight); MOD 6 QUIZ, open at 8am	12/03 Review	12/04 HW7 due (midnight) (only graded for completeness) and solutions will be posted soon after
FINAL EXAM WEEK		12/08		**12/10, open at 8am, 24-hr window; 2.5 hr to complete**	

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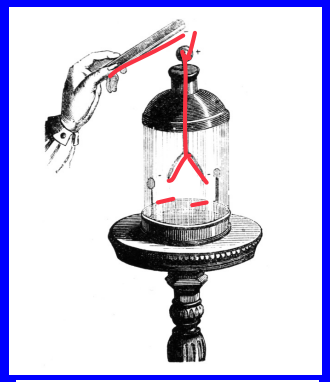
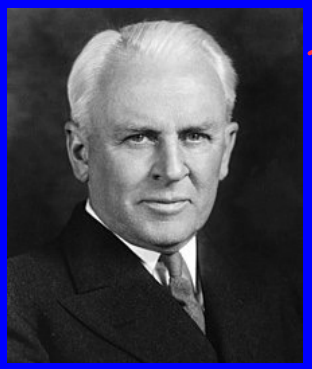
Radiation Belt Dynamics + Sources

Xinlin Li

In early 1900s scientists had been puzzled by the levels of ionizing radiation measured in the atmosphere and on Earth

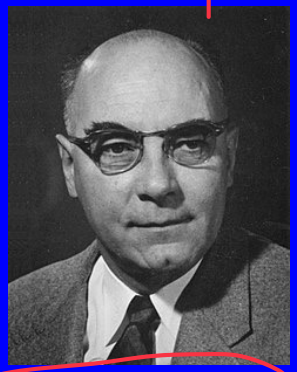


Gold-leaf electroscope

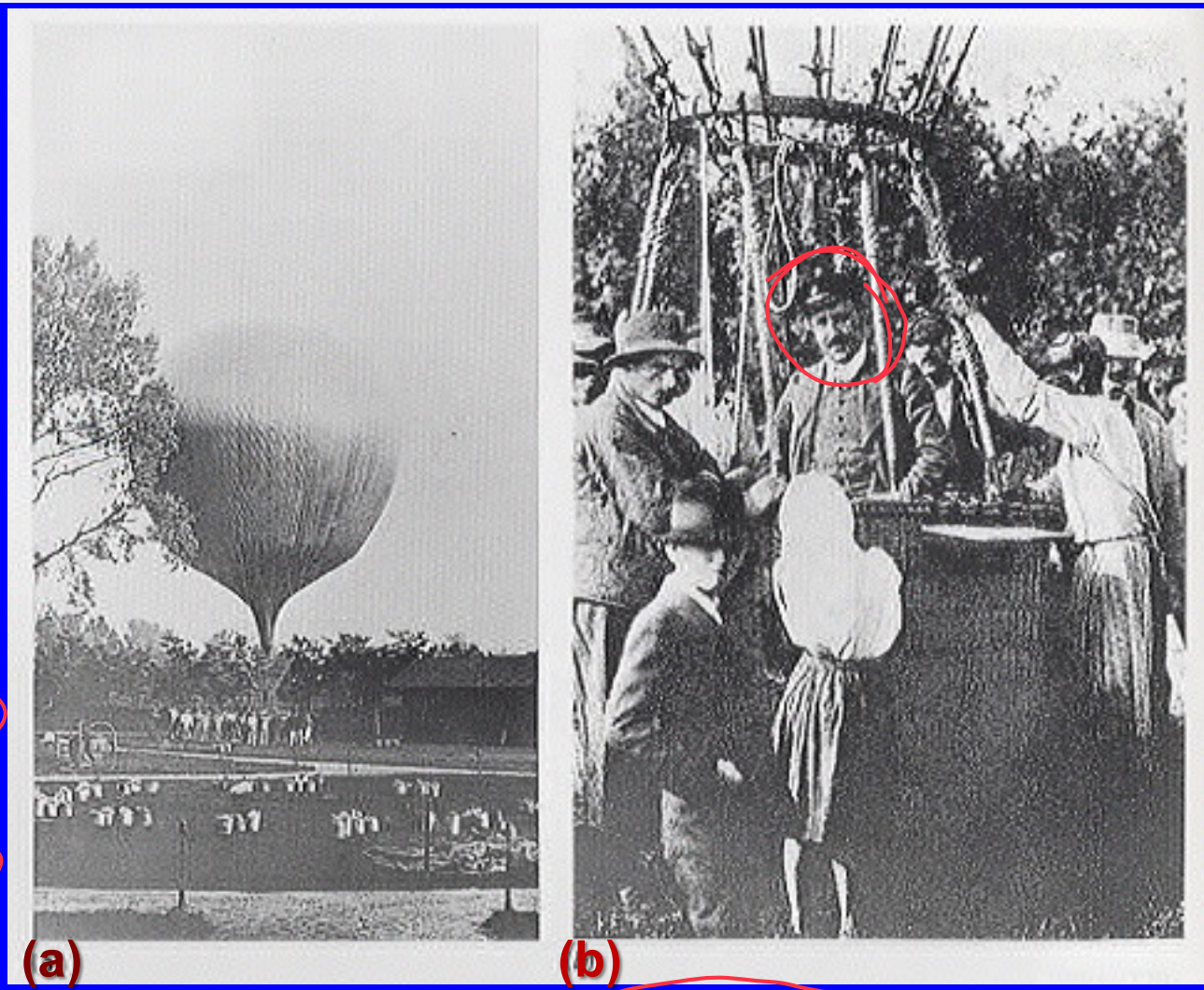


R. A. Millikan

Millikan proved that this radiation is indeed of extraterrestrial origin, and named it "cosmic rays" in 1925.



C. D. Anderson



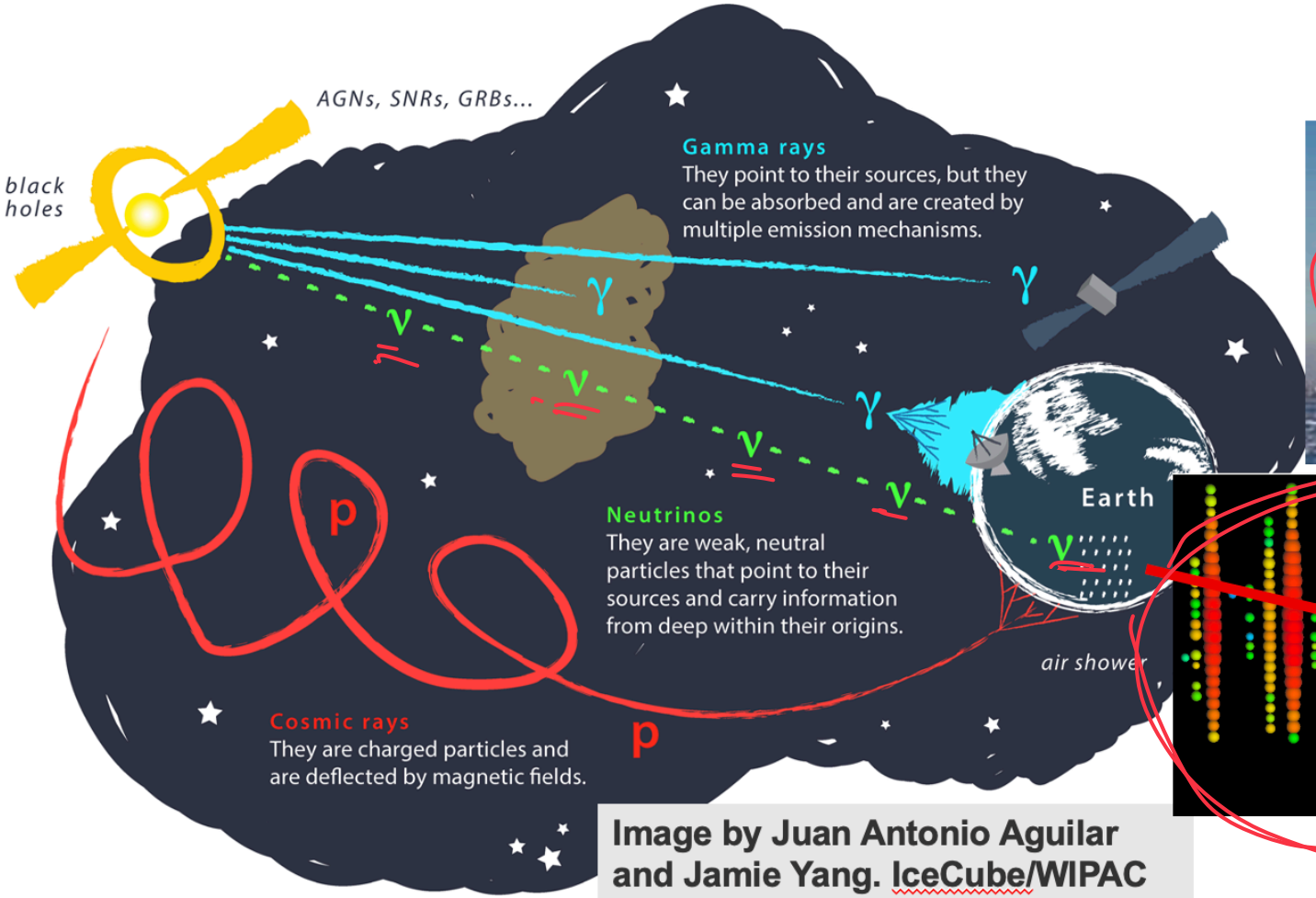
(a) (b)

The balloon flights of Victor Hess
(a) Preparation of one of his flights in 1911-12
(b) Hess after a successful Balloon flight

Positron and muon were discovered in cosmic rays by Anderson.

Hess and Anderson shared the 1936 Nobel Prize in Physics

The origins of cosmic rays are still an on-going study. Most of them are protons likely produced by supernovae in our galaxy, some extreme energetic protons (>10^{15} eV) could have come from other galaxies.



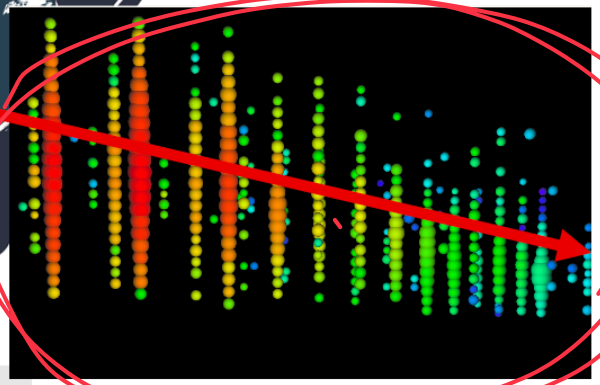
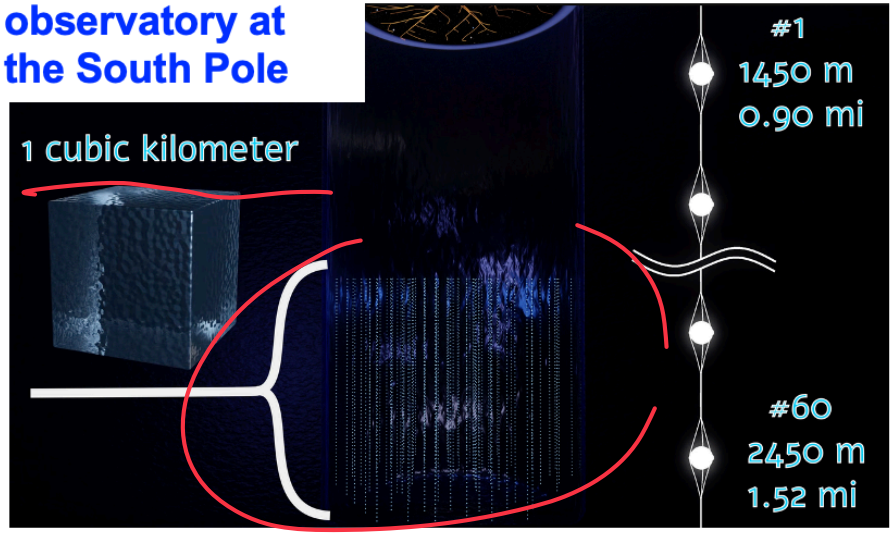


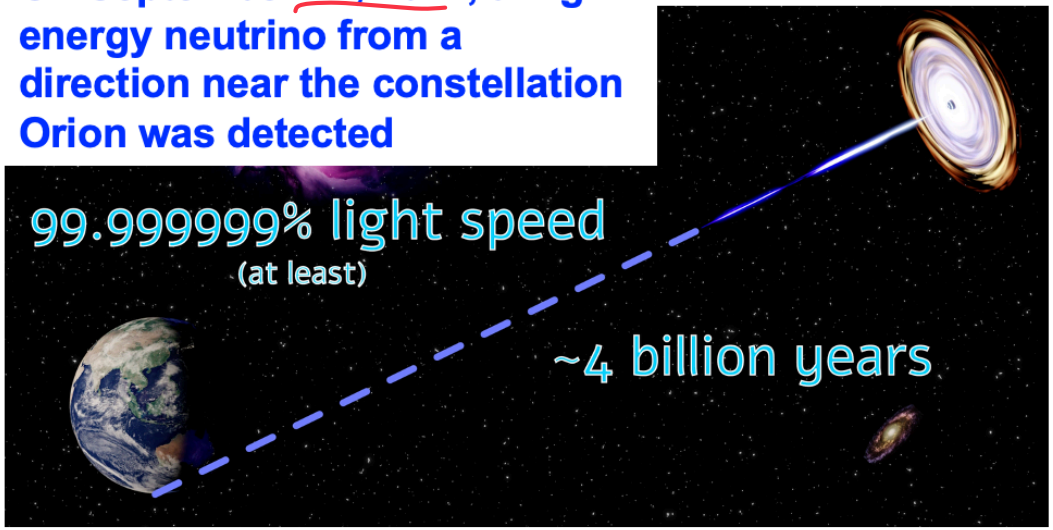
Image by Juan Antonio Aguilar and Jamie Yang. IceCube/WIPAC

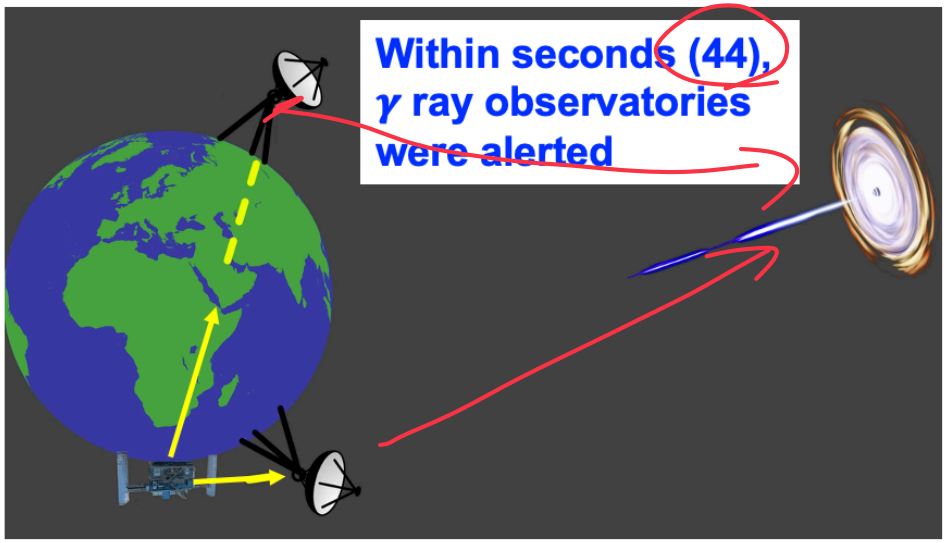
University of Colorado **Boulder**

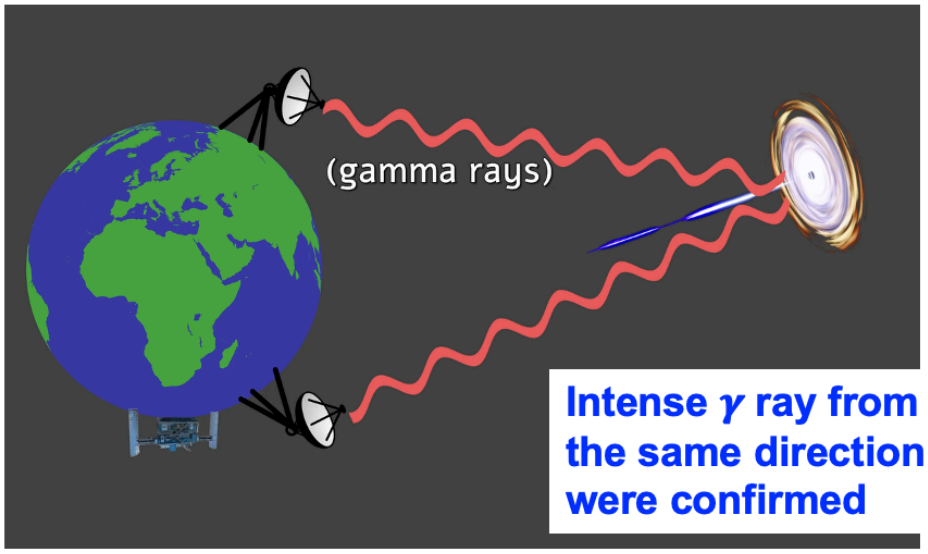
IceCube neutrino observatory at the South Pole



On September 22, 2017, a high energy neutrino from a direction near the constellation Orion was detected







University

LUCA BRUNO — THE ASSOCIATED PRESS

Physicist Francis Halzen flashes a victory sign in his hotel in Turin, Italy, Tuesday, after he was awarded the Nobel Prize in Physics for his contributions to the IceCube Neutrino Observatory and the discovery of high-energy neutrinos of astrophysical origin.

October 6, 2026

Scientist who hunts mysterious ghost particles wins Nobel Prize

By Adithi Ramakrishnan, Stefanie Dazio and Kostya Manenkov
The Associated Press

STOCKHOLM » Belgian-born scientist Francis Halzen won the Nobel Prize in physics Tuesday for his efforts the construction of the IceCube Neutrino Observatory in Antarctica, which glimpsed the first high-energy neutrinos from faraway galaxies.

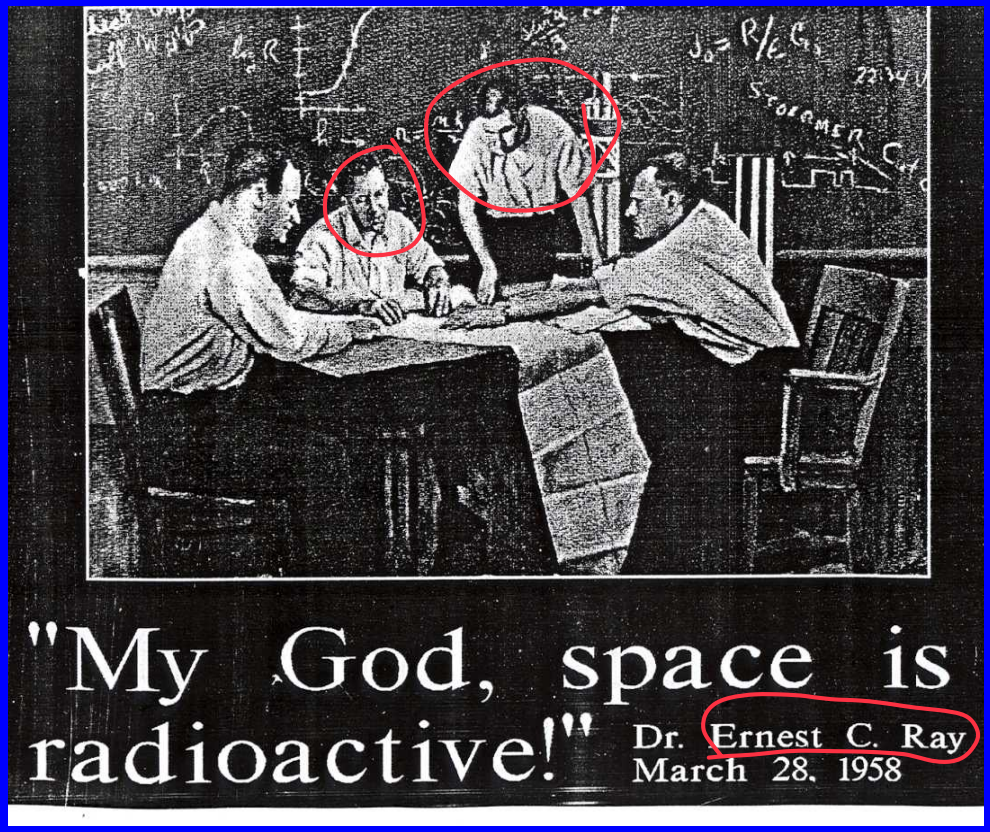
"This experiment in the Antarctic is a revolutionary way of understanding the universe that we didn't quite like Francis Halzen, whose idea to create a neutrino detector under almost a mile of ice has led to a remarkable multinational and multi-institutional scientific collaboration and a fundamental shift in how we think about the universe."

Sputnik-I, 1957-10-4, Soviet Union (22 day, 215x939, 65.1°, 83.6 kg)

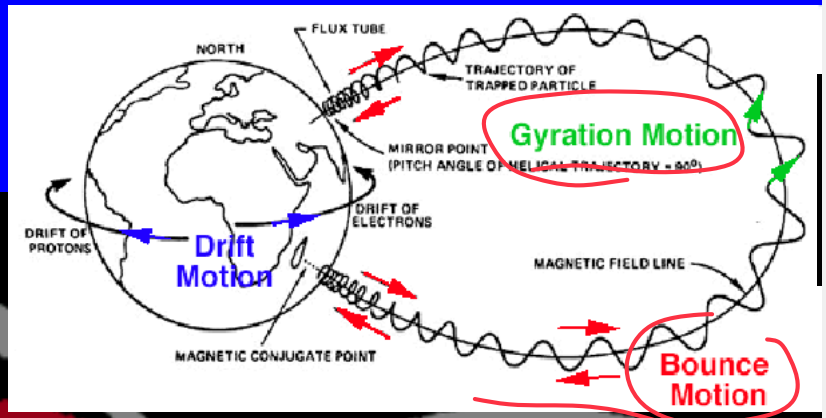
Explore I, 1958-1-31, USA (111 day, 358x2550, 33.24°, 14kg)

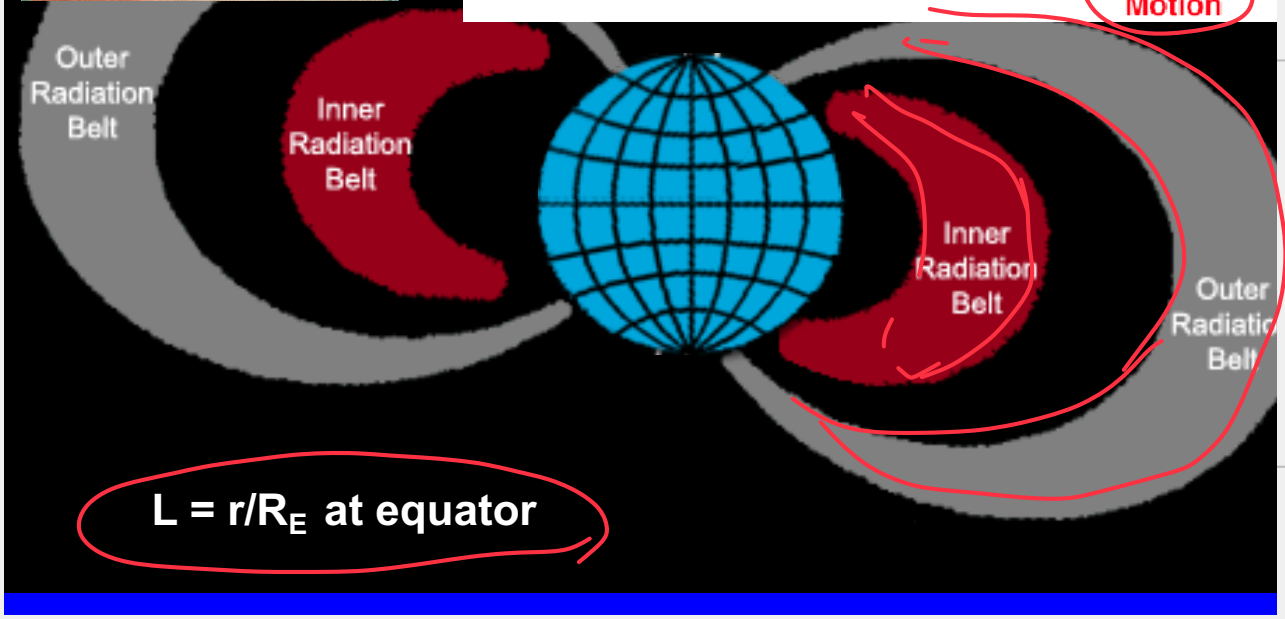
Celebration of the success of Explorer I

Prelude to the Discovery of the Radiation Belts

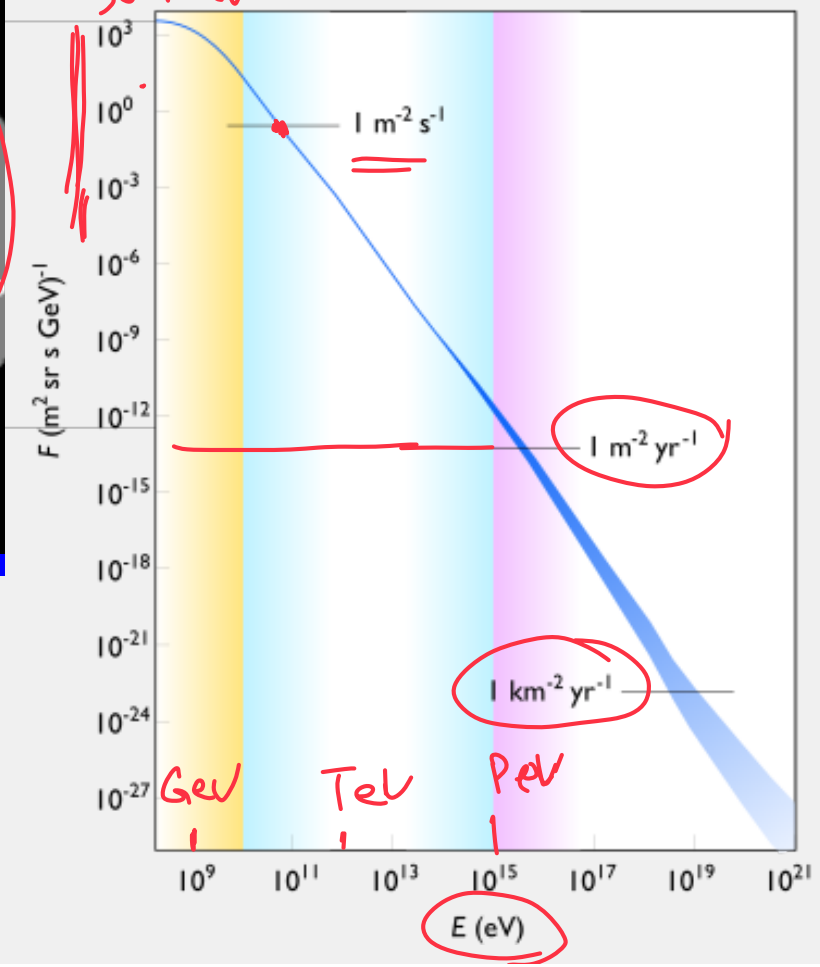


James Van Allen (May 1959)





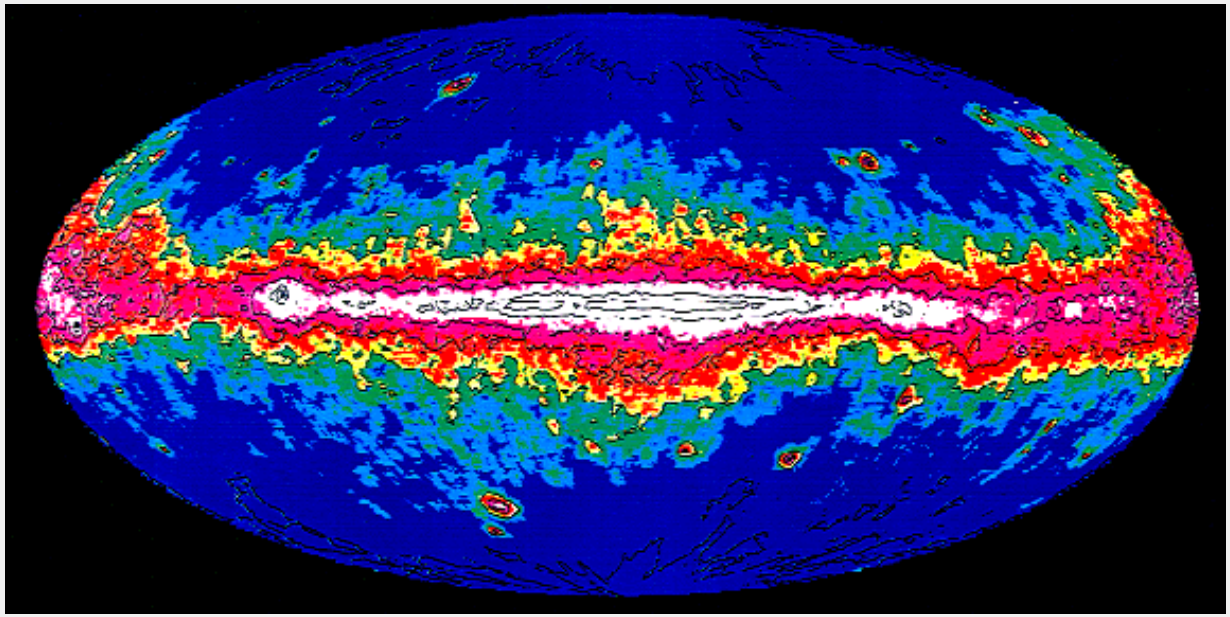
- **CRAND: Cosmic Ray Albedo Neutron Decay**
  - Cosmic rays are mostly protons between 100 MeV (10^8 eV) and 10 GeV (10^{10} eV), with long, high energy tail
  - Average Flux at top of the atmosphere is about 10^4 particles/m²/sec



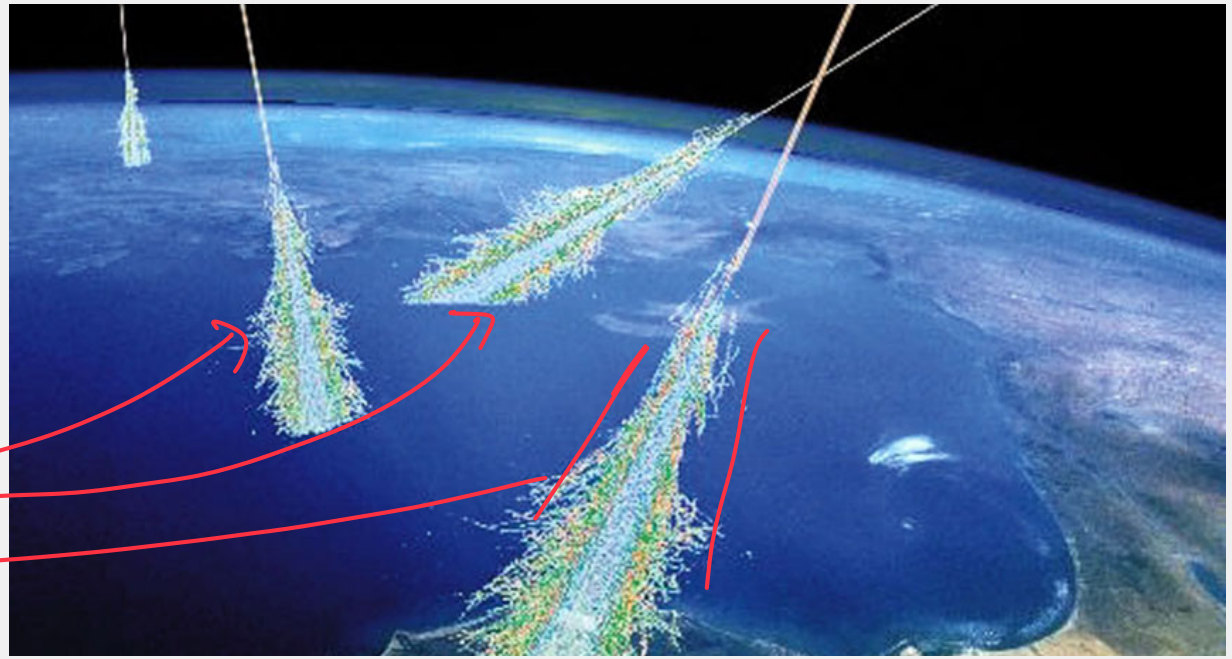
Galactic Cosmic Rays

- **First**: cosmic rays are not RAYS (like gamma rays); they are PARTICLES*
- 90% high-energy protons
- 9% He^{++} (alpha particles),
- 1% heavier nuclei

(*also some gamma rays)

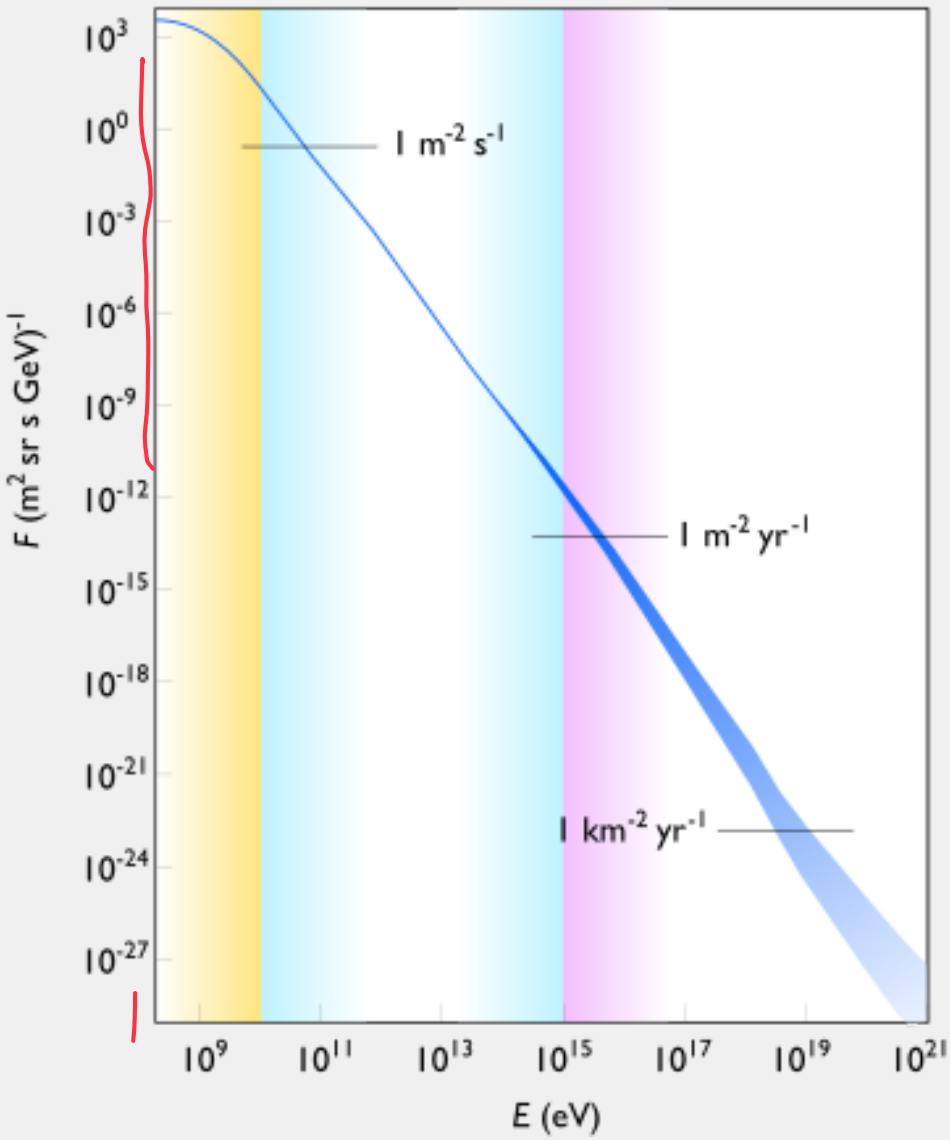


- Originate primarily from within our galaxy
- Particles accelerated by supernova *remnants*
- Radiation at Earth's surface created by *cosmic ray showers*

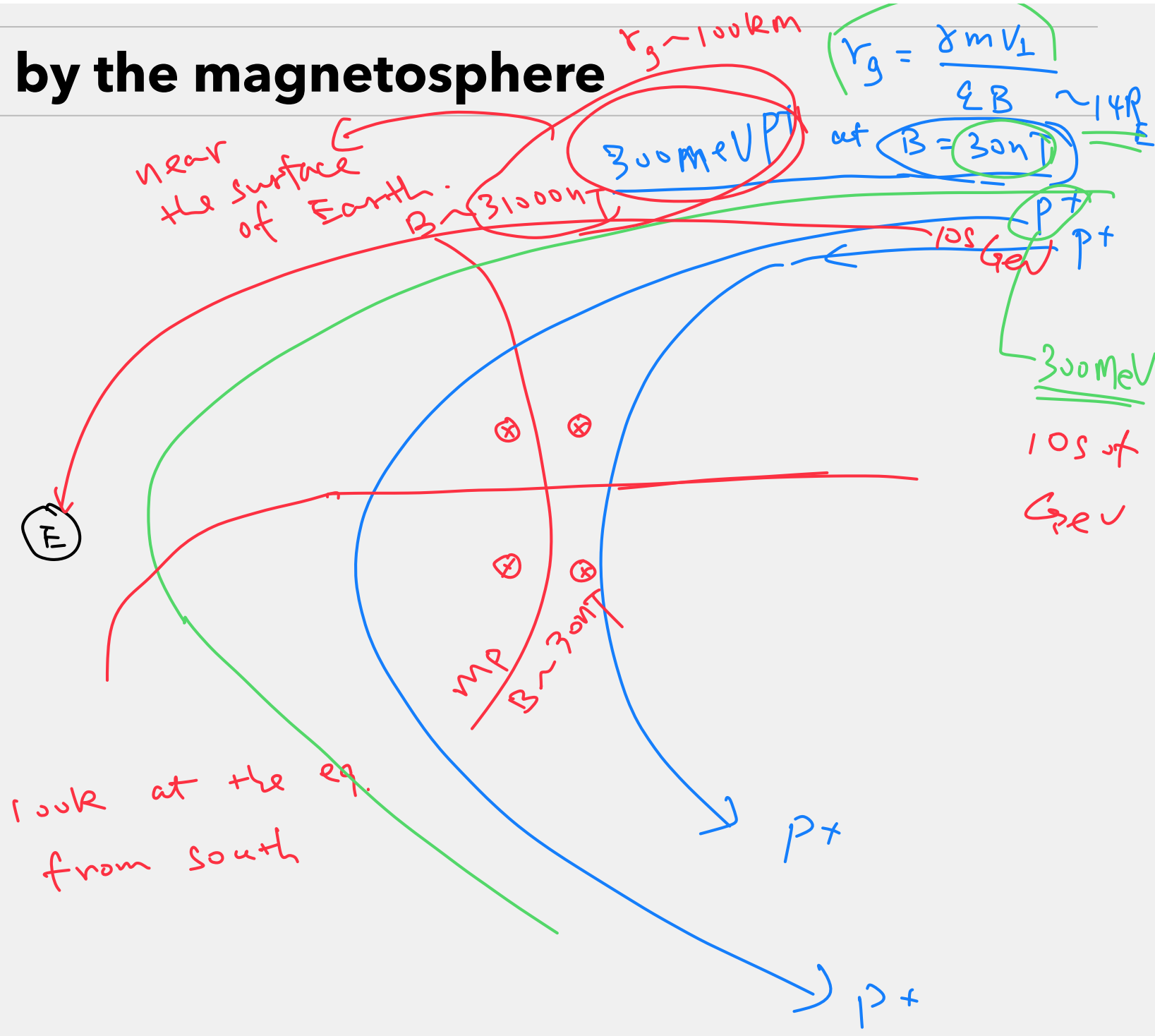


Cosmic Ray Flux

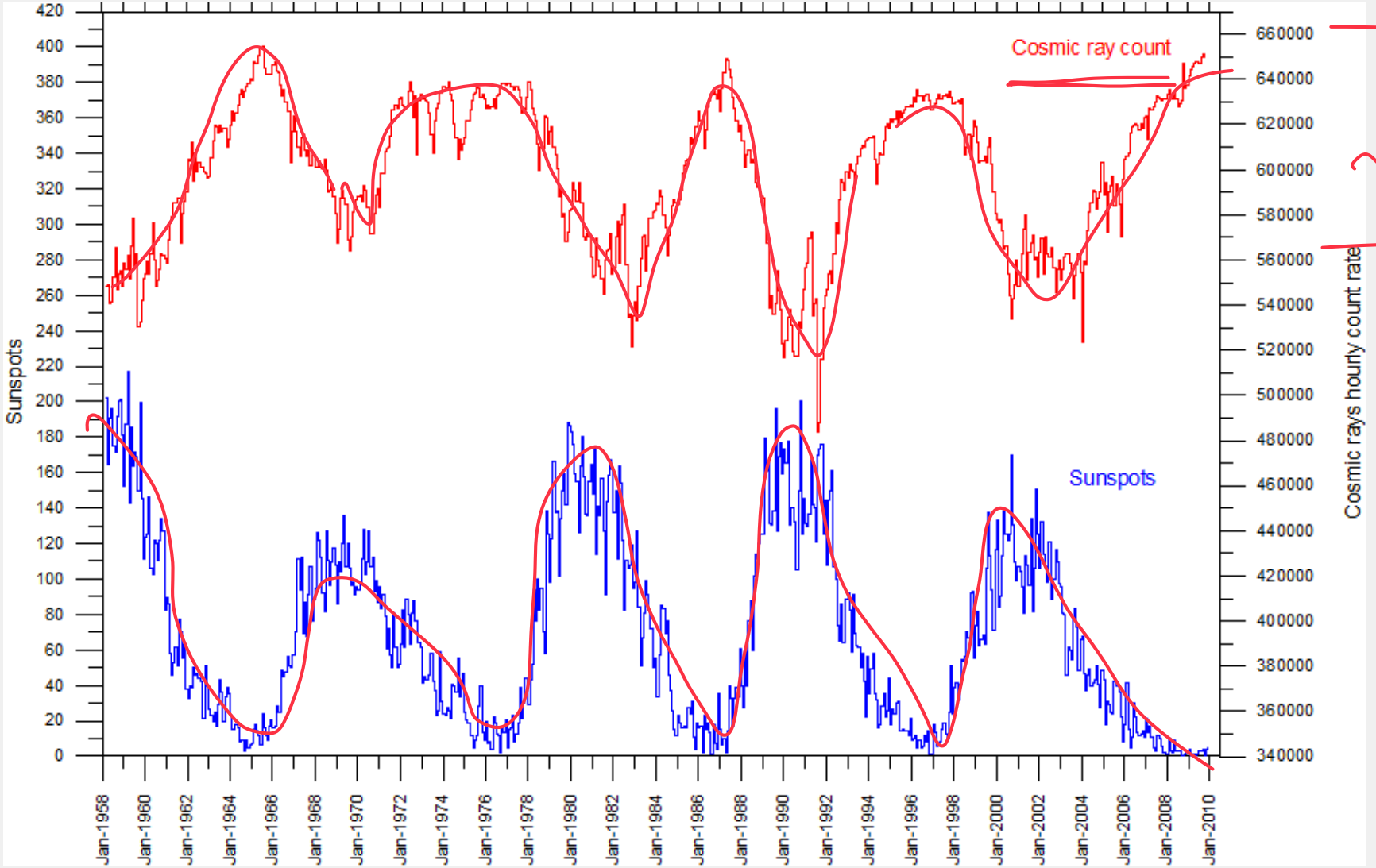
- Vast majority of GCRs have energies in ~GeV range (10^9 eV); peak at <~0.3 GeV
- Lower energies are deflected by IMF and magnetosphere
- Exponential distribution of GCR flux for higher energies
- The effect of a particle with energies above 10^{17} eV will be directly measurable at the Earth's surface
- Highest energy observed: 3 x 10^{20} eV; called the "Oh-My-God" particle (likely coming from other galaxies)
  - For comparison, LHC (Large Hadron Collider) can accelerate up to 7 TeV or 7 x 10^{12} eV



Deflection by the magnetosphere



Cosmic Rays and the Solar Cycle



- Interplanetary magnetic field (IMF) provides some shielding from GCRs
- During solar minimum, interplanetary magnetic field is **weakest**, so flux is highest

IMF weaker during solar min
~15 → 20%

Cosmic Rays and the Solar Cycle

protons measured by Voyager-1 & -2 as they were heading out our heliosphere

Galactic Cosmic Ray

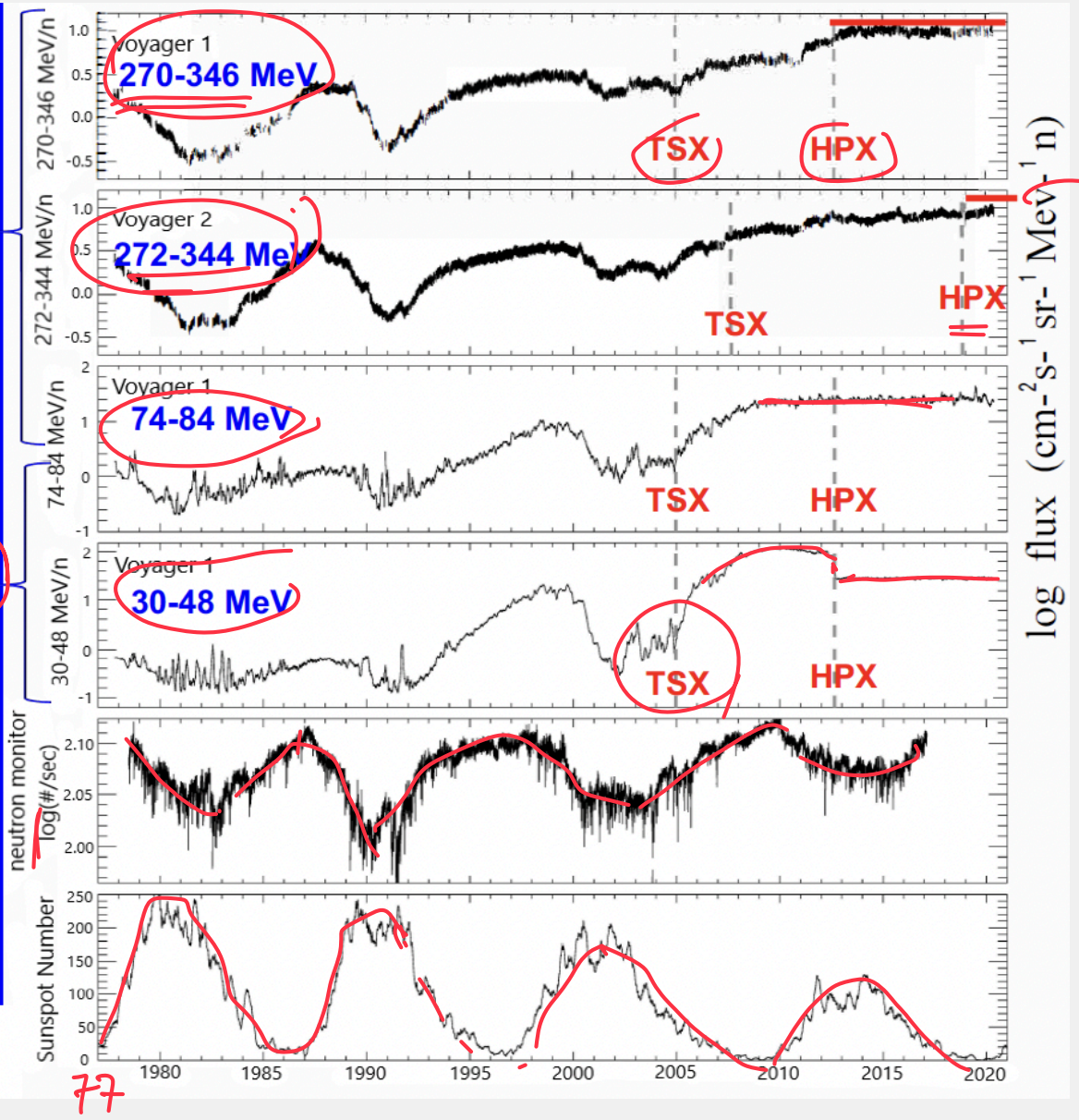
Different behavior for different energy protons

Anomalous Cosmic Ray

Ground Neutron Monitor Measurements (Greenland)

Sunspot Number

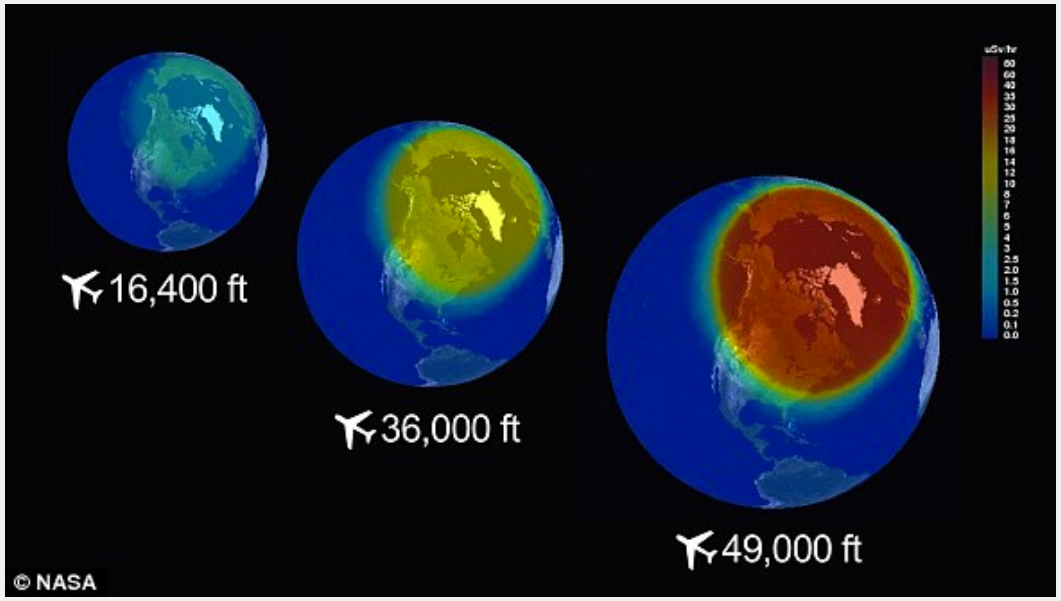
(Extended Figure 1 of Li et al., JGR, 2020)

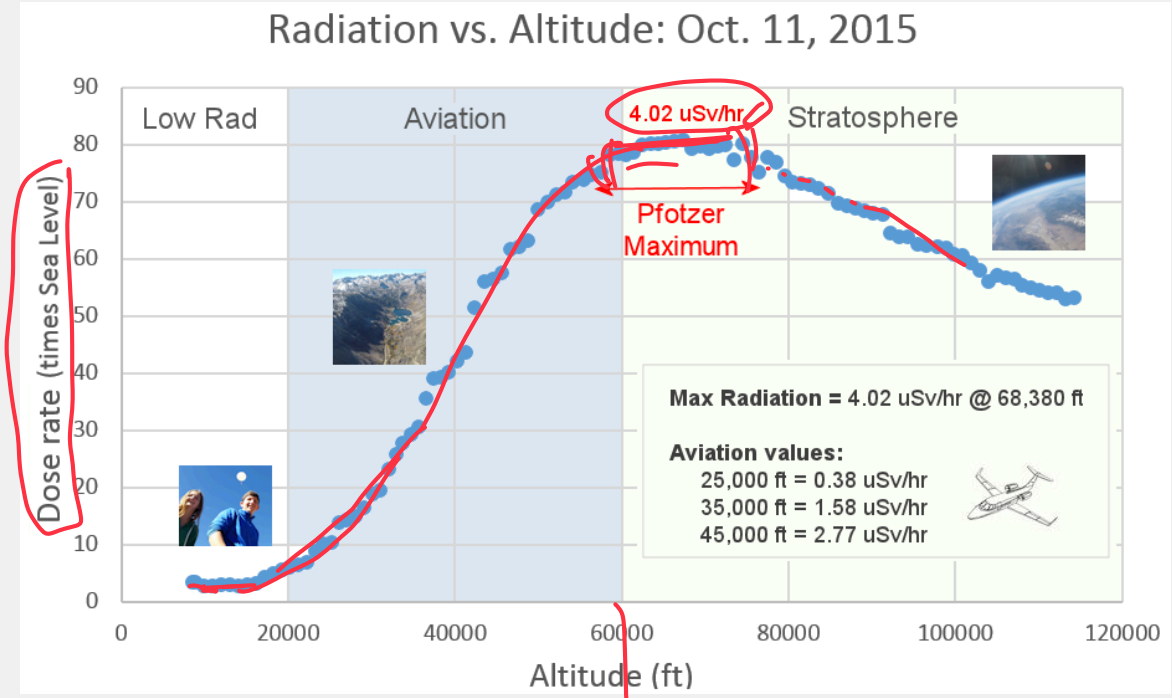


- Interplanetary magnetic field (IMF) provides some shielding from GCRs
- During solar minimum, interplanetary magnetic field is **weakest**, so flux is highest

Where do Cosmic Rays have most impact?

- Highest energy cosmic rays can reach the atmosphere anywhere on Earth: they penetrate right through the magnetosphere
- Lower energies (higher fluxes!) are deflected by magnetosphere, but can penetrate in the polar regions, same as SEPs
- Peak of energy deposition is at 60,000 ft altitude: Pfotzer maximum
- aviation industry crew are exposed to 2x higher radiation on average compared to us surface-dwellers





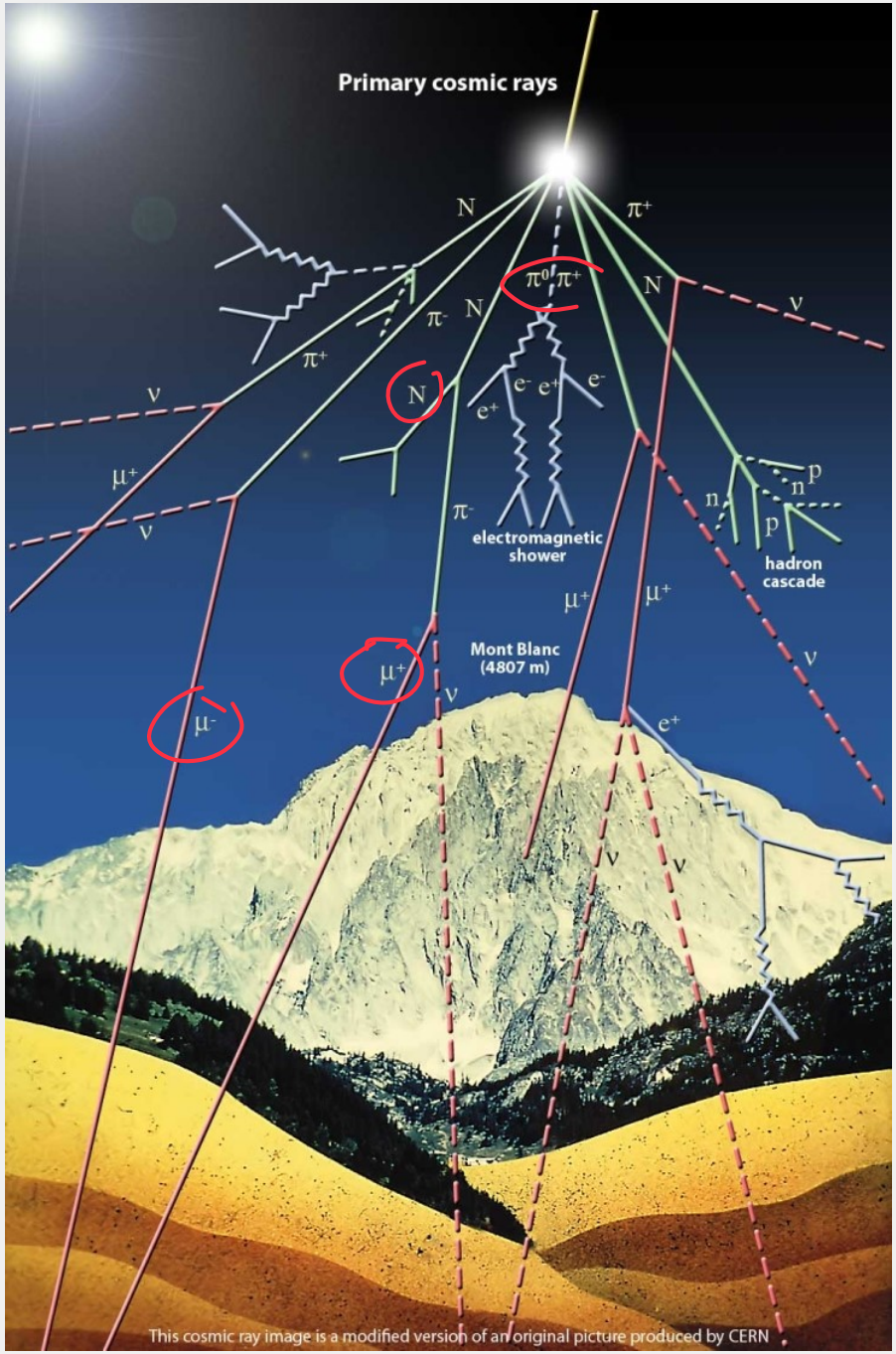
GCR effects in the atmosphere

- As GCR particles reach the atmosphere, they interact with atmospheric molecules (N2 and O2)
- Due to high energies, the ensuing particle physics is complicated!
  - pions, muons, neutrinos, neutrons, protons, electrons, antimatter, etc.
  - Muons (207 of e- mass) are easily detectable on the ground. Used for REPTile' and REPTile-2's initial tests

Detector	Measured	St. Dev.	95% Conf.	Expected
1 (20mm)	1.07	1.03	0.02	~1.2
2 (40mm)	3.67	1.93	0.04	~4.7
3 (40mm)	3.47	1.85	0.04	~4.7
4 (40mm)	3.42	1.85	0.04	~4.7

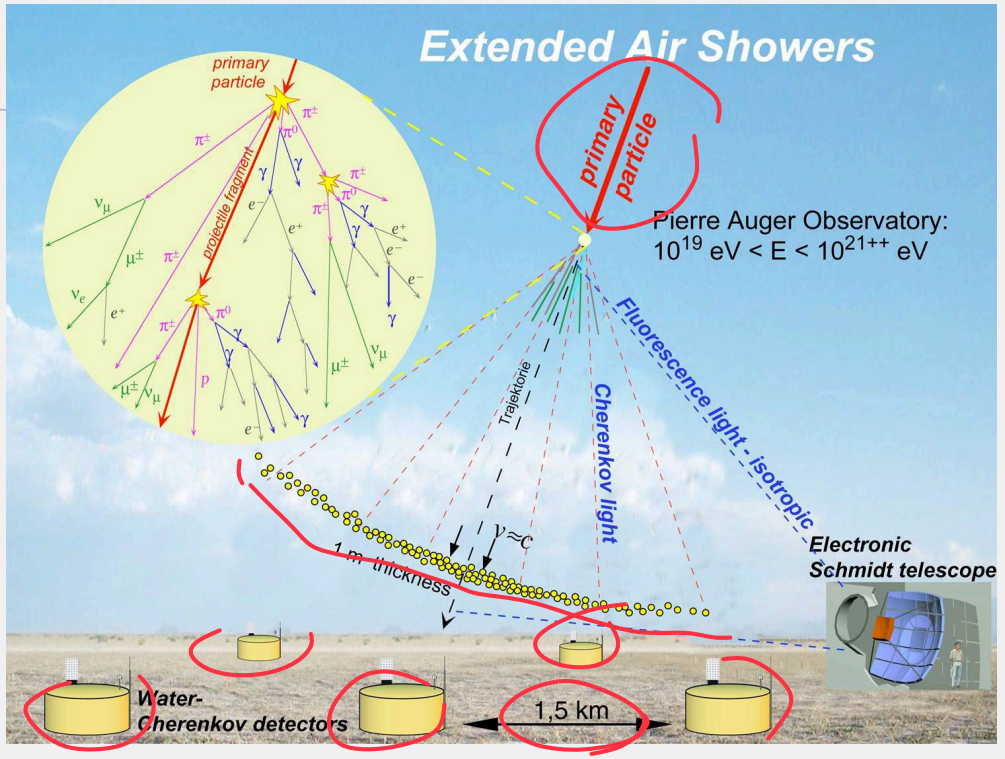
Table 6: Averaged Muon Countrates (#/6s)

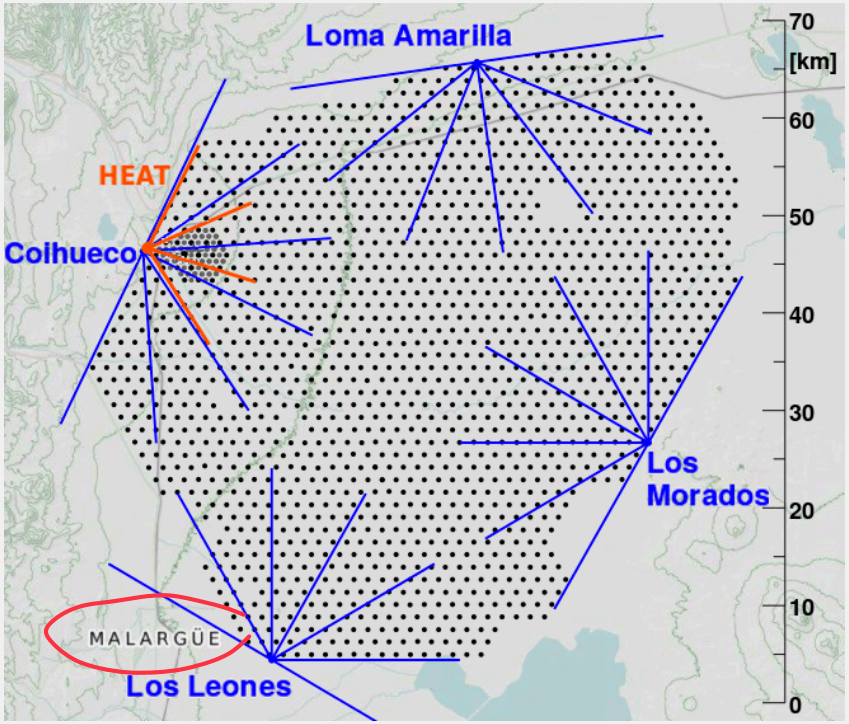
- **Called a "cosmic ray air shower"**
- We are constantly exposed to these, with a relatively radiation dose



Pierre Auger Observatory

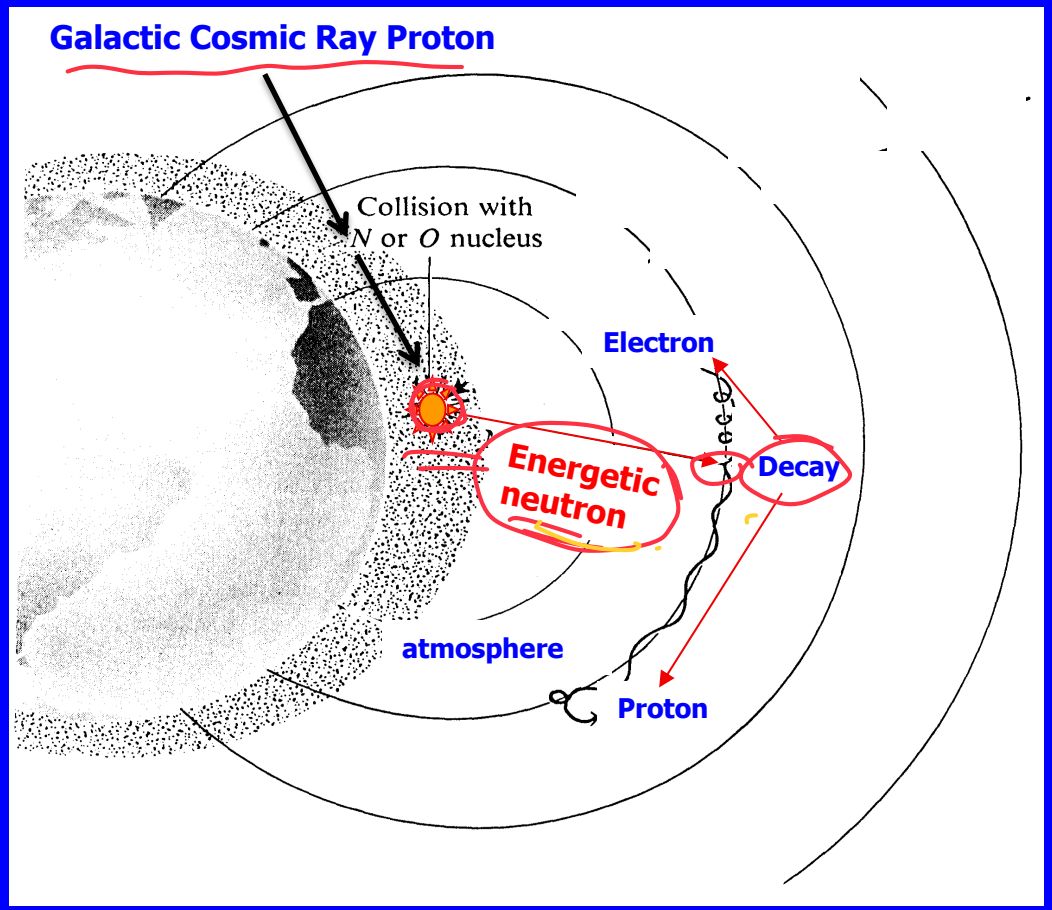
- Cosmic Rays regularly studied from the ground
  - "Surface detectors": tasks of 3000 gallons of pure water to detect muons; create Cherenkov radiation (light) when passing through water
  - "Fluorescence detectors": sensitive optical detectors to measure flash of light in the sky during air showers





Sources for Inner Belt protons: CRAND and SEPs

Cosmic Ray Albedo Neutron Decay (CRAND) contributions to the inner belt protons (and electrons)



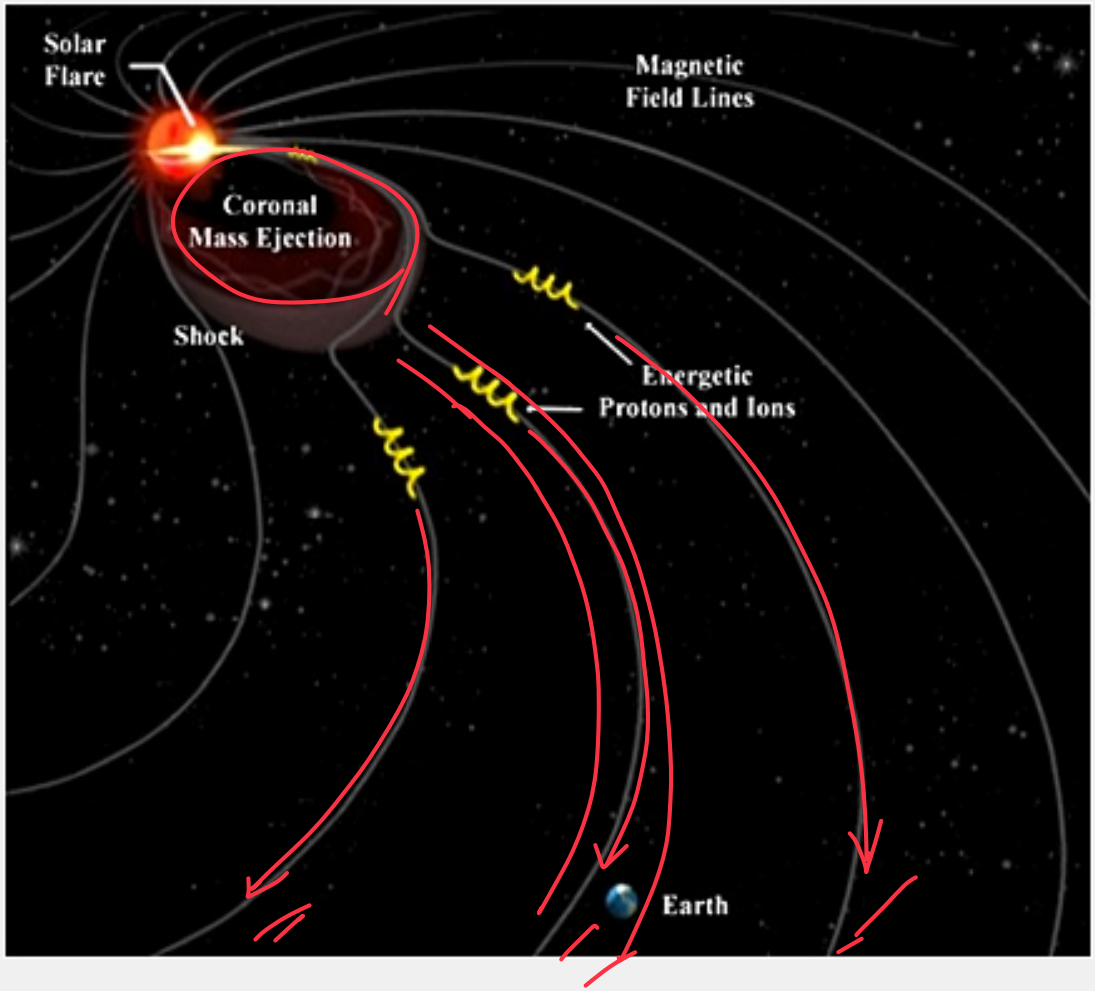
Fast neutrons made by direct interaction of high-energy cosmic rays: knock-on neutrons

proton + electron + antineutrino (production of ≥10 MeV energetic protons (L ~ 1.5) that can be trapped).

Some energetic neutrons can also reach at the surface of the Earth, detected by Neutron Monitors

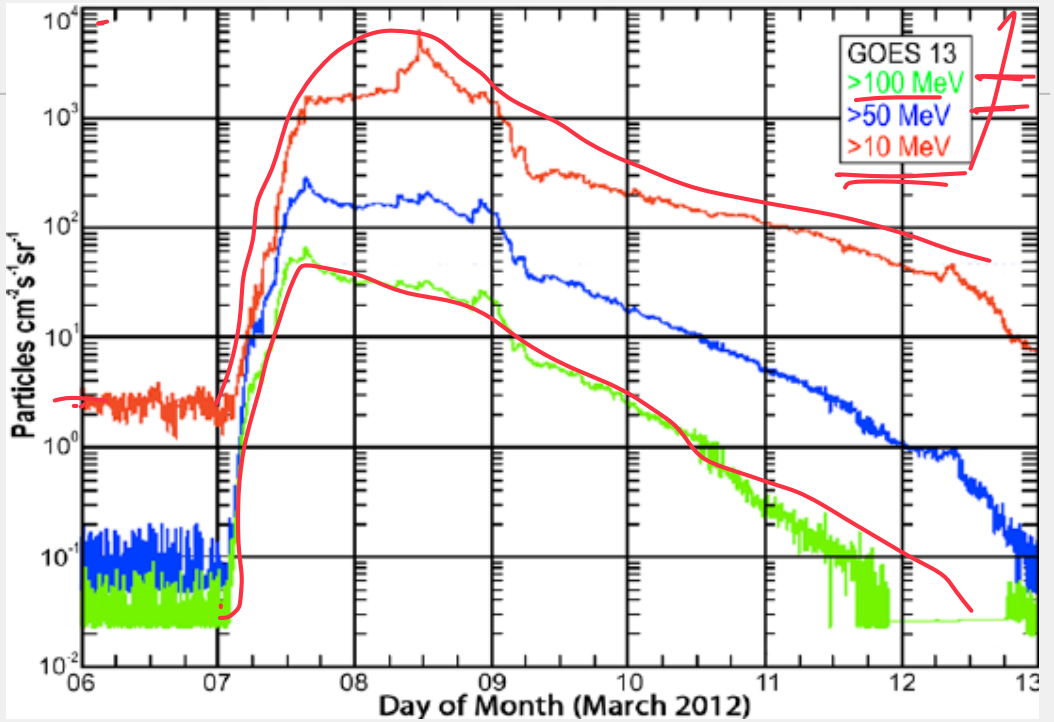
Solar Energetic Particles

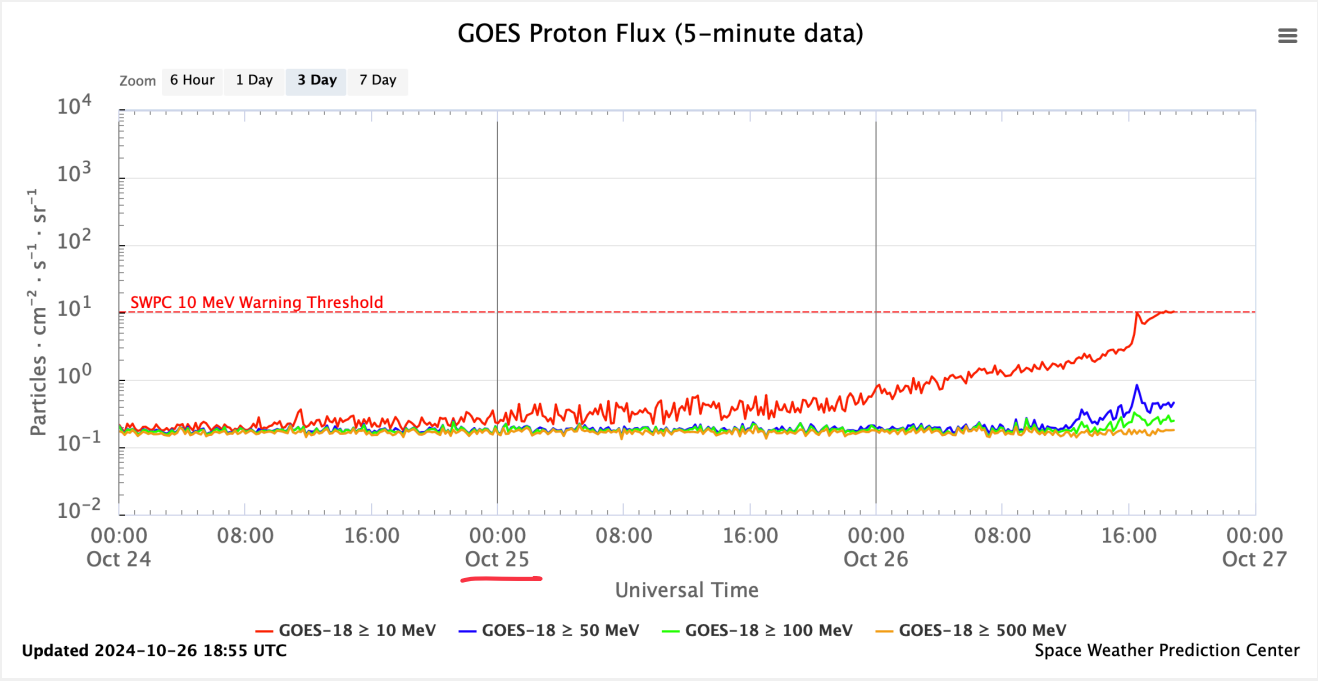
- High-energy particles coming from the sun; protons, electrons, and other heavier particles
- tens of keV to many GeV
- highest energies can reach Earth in a few minutes; forecasting is difficult!

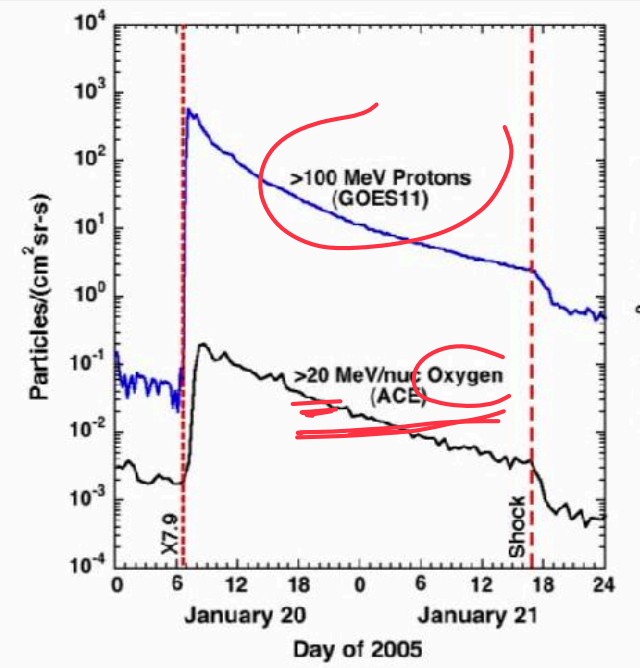


SEP Fluxes

- Background fluxes of 0.1 particles/cm²/sec/str (depending on energy)
- SEP event fluxes increase 1000x or more
- Increased fluxes can persist for days
- Over duration of SEP event, fluence* over 1 MeV can reach 10^9 particles/cm²

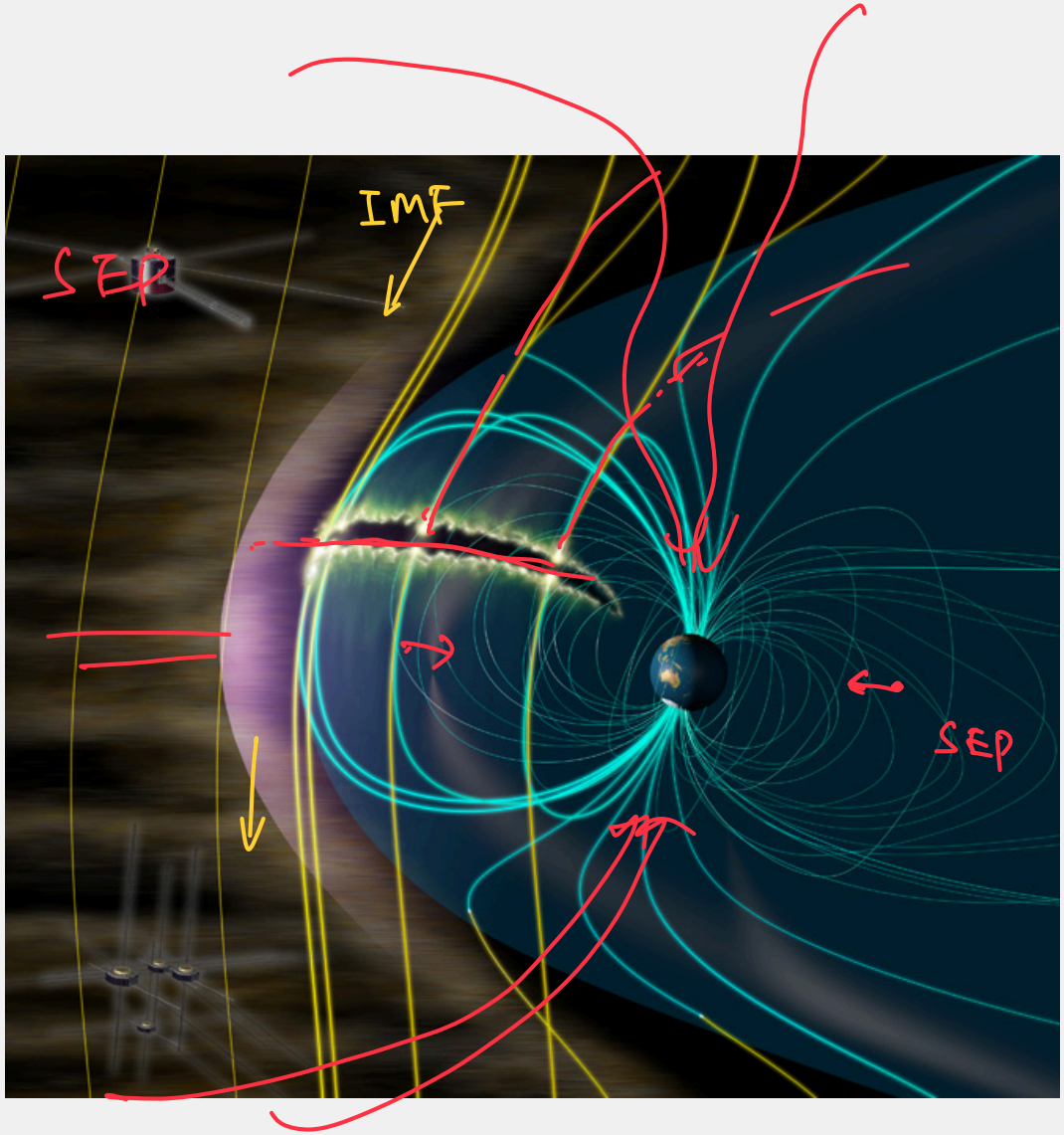






SEP entry into Magnetosphere

- SEPs can enter the magnetosphere where there are "open" magnetic field lines: Polar cap region
  - Lead to polar cap absorption
  - Radiation hazard in polar regions!
- During storms, magnetosphere is compressed; polar cap extends to lower latitudes
- Some SEP can be trapped and transported inward and merge with the inner proton belt
- **During storms, SEPs can affect spacecraft at lower latitudes**



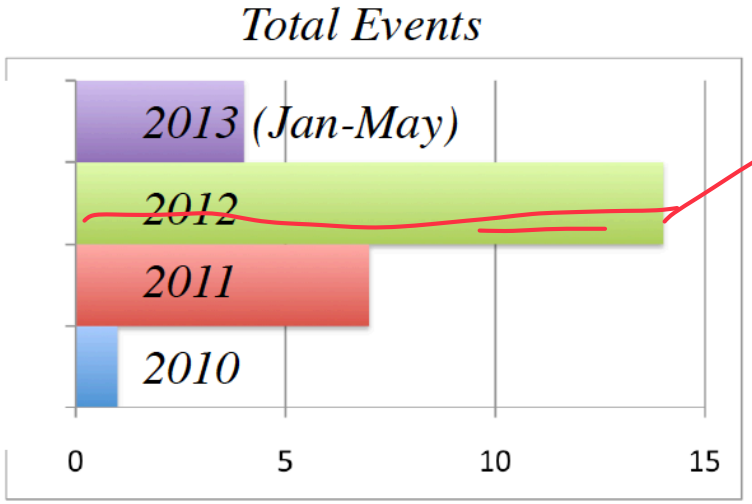
NOAA SEP Categorization

NOAA Space Weather Scale for Solar Radiation Storms

Category		Effect	Physical measure	Average Frequency (1 cycle = 11 years)
Scale	Descriptor	Duration of event will influence severity of effects		
Solar Radiation Storms			Flux level of >= 10 MeV particles (ions)*	Number of events when flux level was met (number of storm days**)
S 5	Extreme	**Biological:** unavoidable high radiation hazard to astronauts on EVA (extra-vehicular activity); passengers and crew in high-flying aircraft at high latitudes may be exposed to radiation risk.*** **Satellite operations:** satellites may be rendered useless, memory impacts can cause loss of control, may cause serious noise in image data, star-trackers may be unable to locate sources; permanent damage to solar panels possible. **Other systems:** complete blackout of HF (high frequency) communications possible through the polar regions, and position errors make navigation operations extremely difficult.	10^5	Fewer than 1 per cycle < 1/solar cycle
S 4	Severe	**Biological:** unavoidable radiation hazard to astronauts on EVA; passengers and crew in high-flying aircraft at high latitudes may be exposed to radiation risk.*** **Satellite operations:** may experience memory device problems and noise on imaging systems; star-tracker problems may cause orientation problems, and solar panel efficiency can be degraded. **Other systems:** blackout of HF radio communications through the polar regions and increased navigation errors over several days are likely.	10^4	3 per cycle 1/3yr
S 3	Strong	**Biological:** radiation hazard avoidance recommended for astronauts on EVA; passengers and crew in high-flying aircraft at high latitudes may be exposed to radiation risk.*** **Satellite operations:** single-event upsets, noise in imaging systems, and slight reduction of efficiency in solar panel are likely. **Other systems:** degraded HF radio propagation through the polar regions and navigation position errors likely.	10^3	10 per cycle 1/yr
S 2	Moderate	**Biological:** passengers and crew in high-flying aircraft at high latitudes may be exposed to elevated radiation risk.*** **Satellite operations:** infrequent single-event upsets possible. **Other systems:** small effects on HF propagation through the polar regions and navigation at polar cap locations possibly affected.	10^2	25 per cycle 2/yr
S 1	Minor	**Biological:** none. **Satellite operations:** none. **Other systems:** minor impacts on HF radio in the polar regions.	10	50 per cycle 5/yr

How often do SEP events occur? (At Earth)

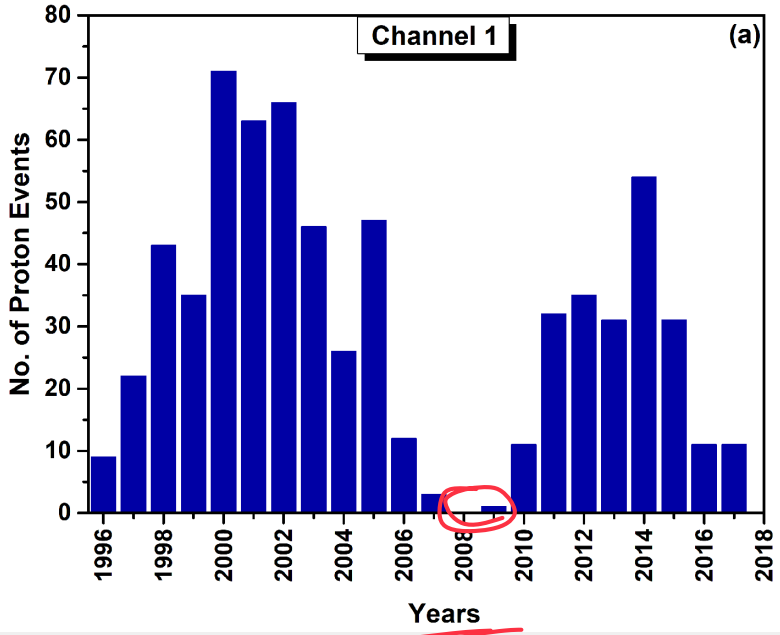
2007-2009: Zero Events - Solar Minimum Indeed!

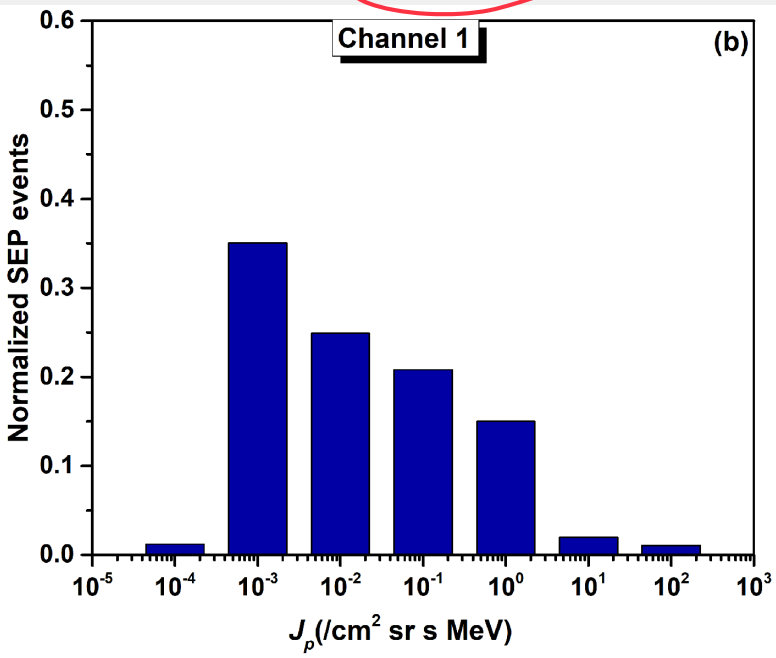


+ 11 yr ~ 2023 Solar Max

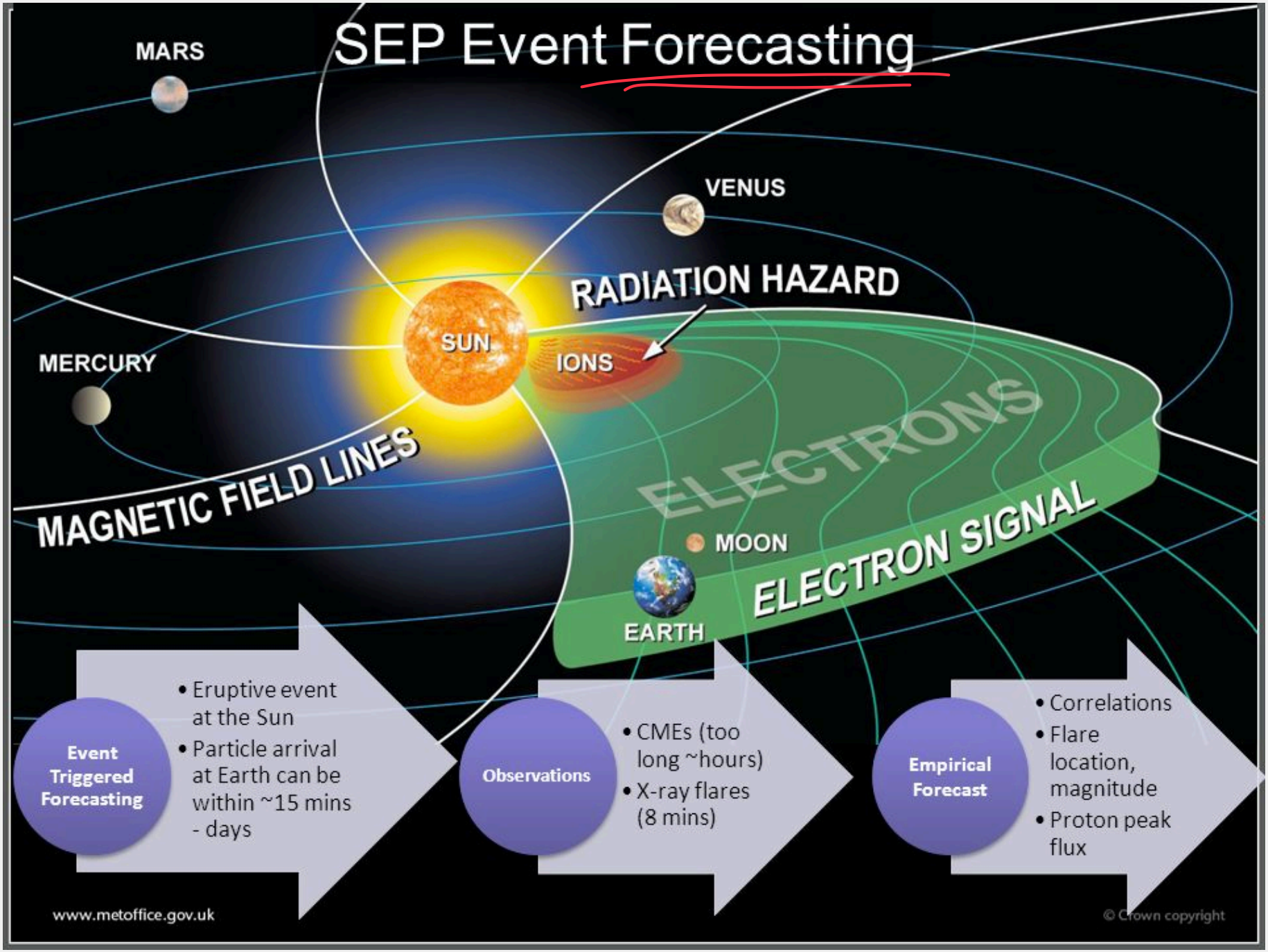
Since March 2011
STEREO A: 16
STEREO B: 11

SOHO ~ L1

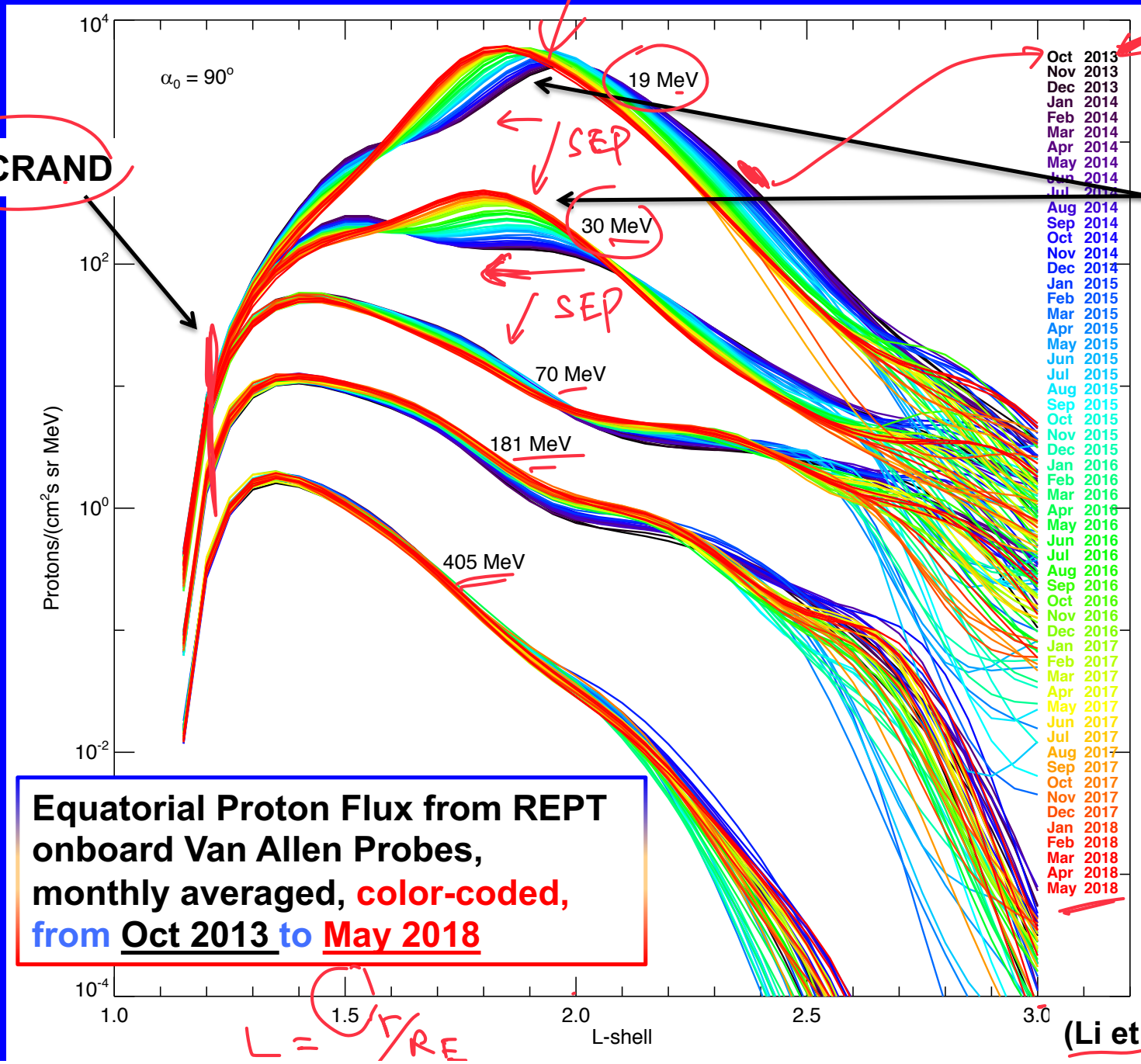




SEP Forecasting: electrons move faster than protons!



Sources for Inner Belt protons: CRAND and SEPs

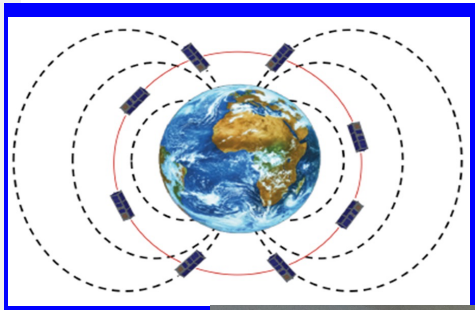


Inward radial diffusion of Solar Energetic Particles (SEP)

Equatorial Proton Flux from REPT onboard Van Allen Probes, monthly averaged, color-coded, from Oct 2013 to May 2018

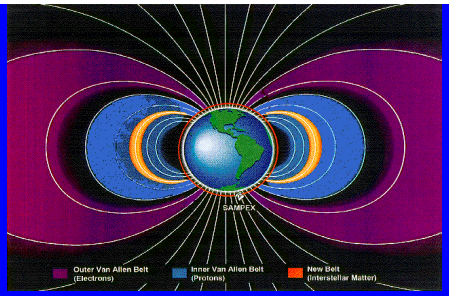
(Li et al., AGU Monogr., 2021)

CRAND is also a source for the inner belt electrons

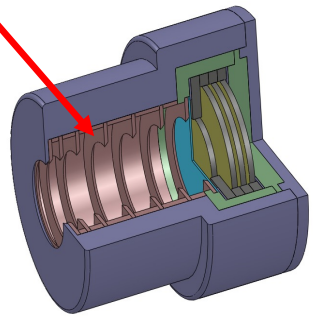


CSSWE: Colorado Student Space Weather Experiment

(Spring of 2010)



$860K



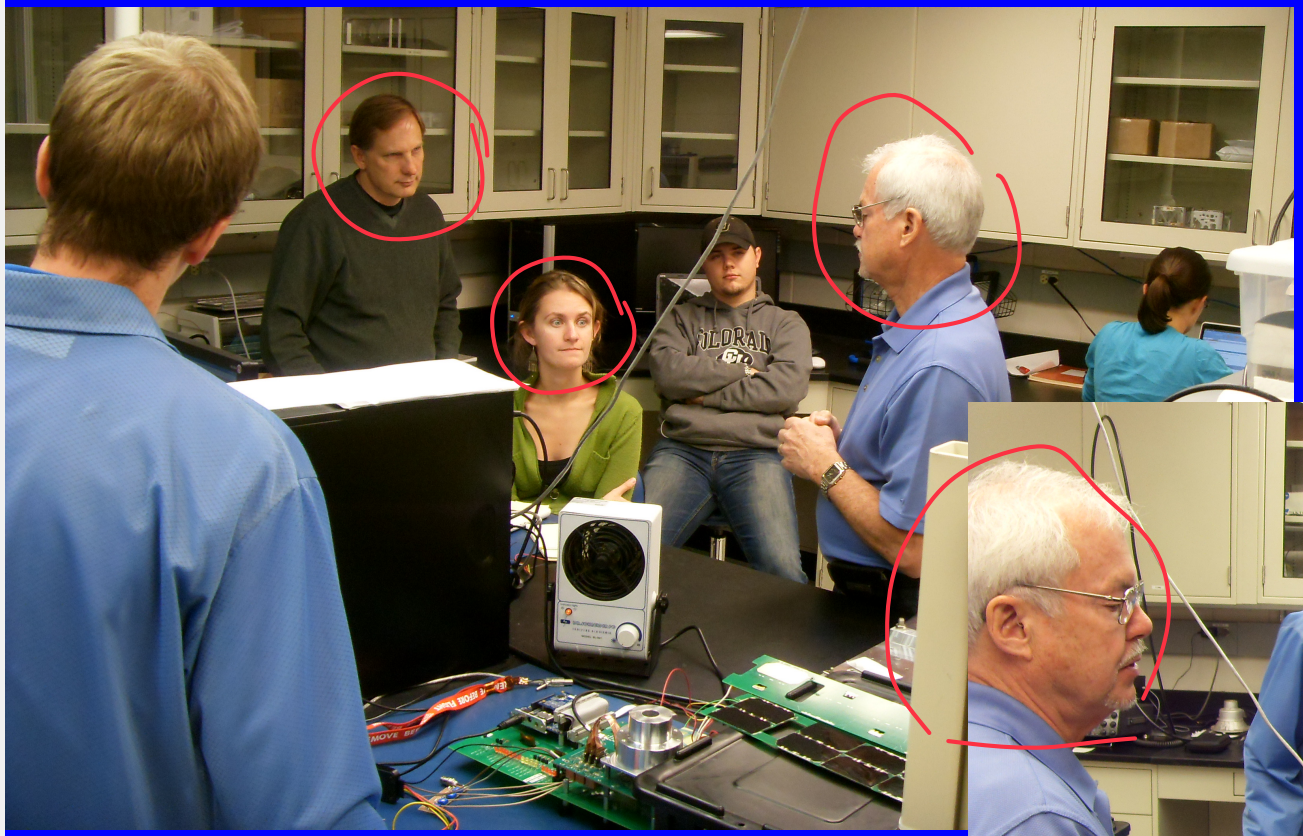
Relativistic Electron-Proton Telescope integrated little experiment (REPTile)

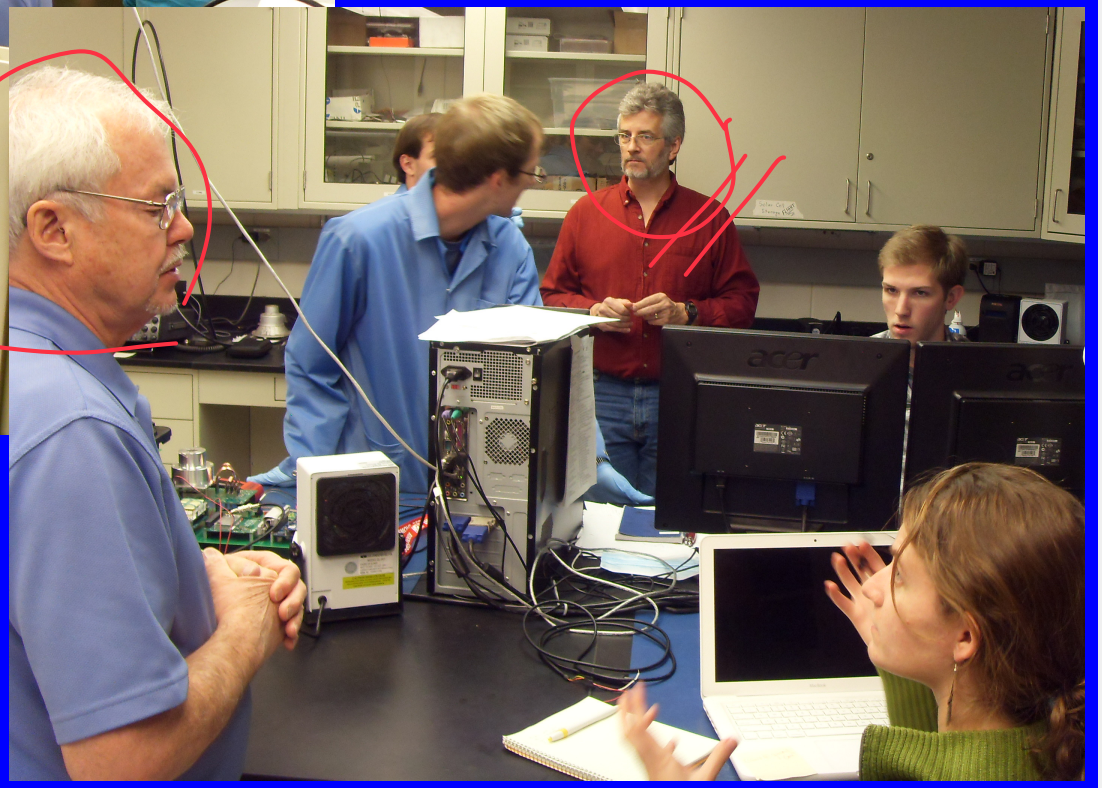
LASP

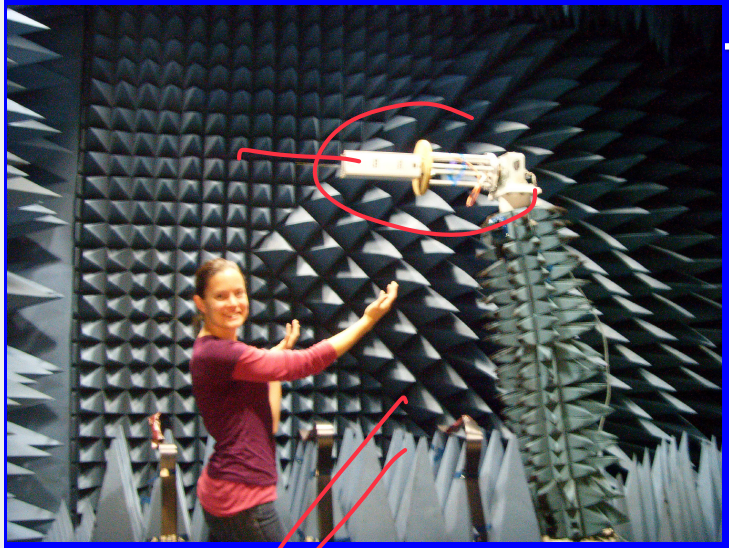
LABORATORY FOR ATMOSPHERIC AND SPACE PHYSICS

Engineering

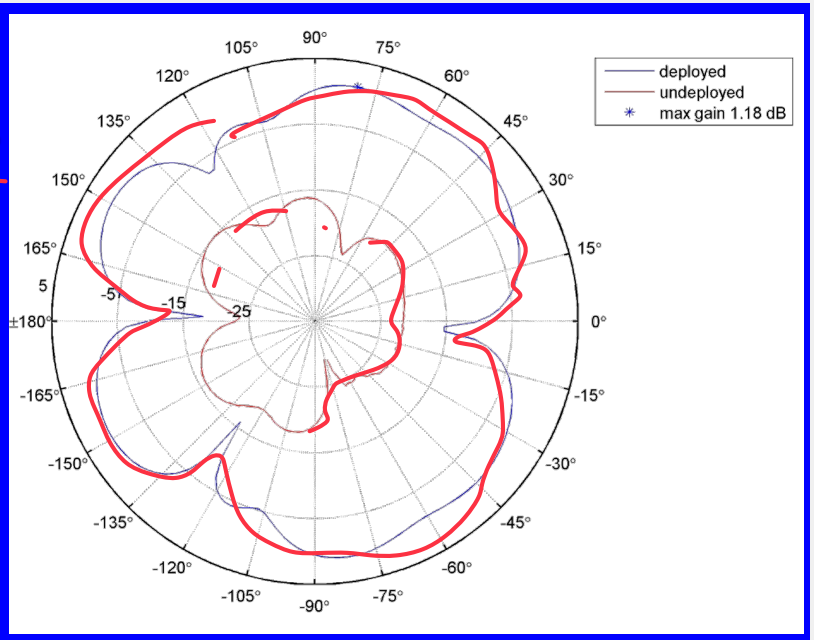
Communication System Debugging, stressful time in the CubeSat Lab (Oct 2011)







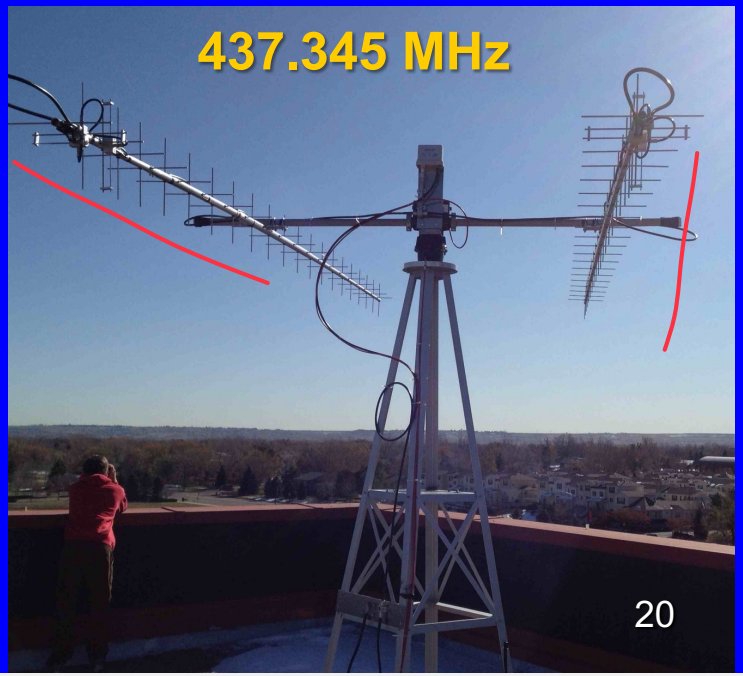
Testing in anechoic chamber of FirstRF Corp. to determine the antenna gain pattern ➔



1st Plugs-out Test (11/10-11/2011):

2nd Plugs-out Test (after Vibe test): 11/29/2011

3rd Plugs-out Test (after T-V test): 12/21/2011



So far, 29 peer-reviewed papers published, → **including two *Nature* papers**

+ 5 Ph.D. Theses Completed
(Drs. Drew Turner, David Gerhardt, Lauren Blum, Quintin Schiller, and Kun Zhang)

Directly involved >65 grad and undergrad students

Resolved a 60-year mystery of the source population of inner radiation belt electrons based on CSSWE/REPTile measurements.

Publication List Associated with CSSWE CubeSat Mission

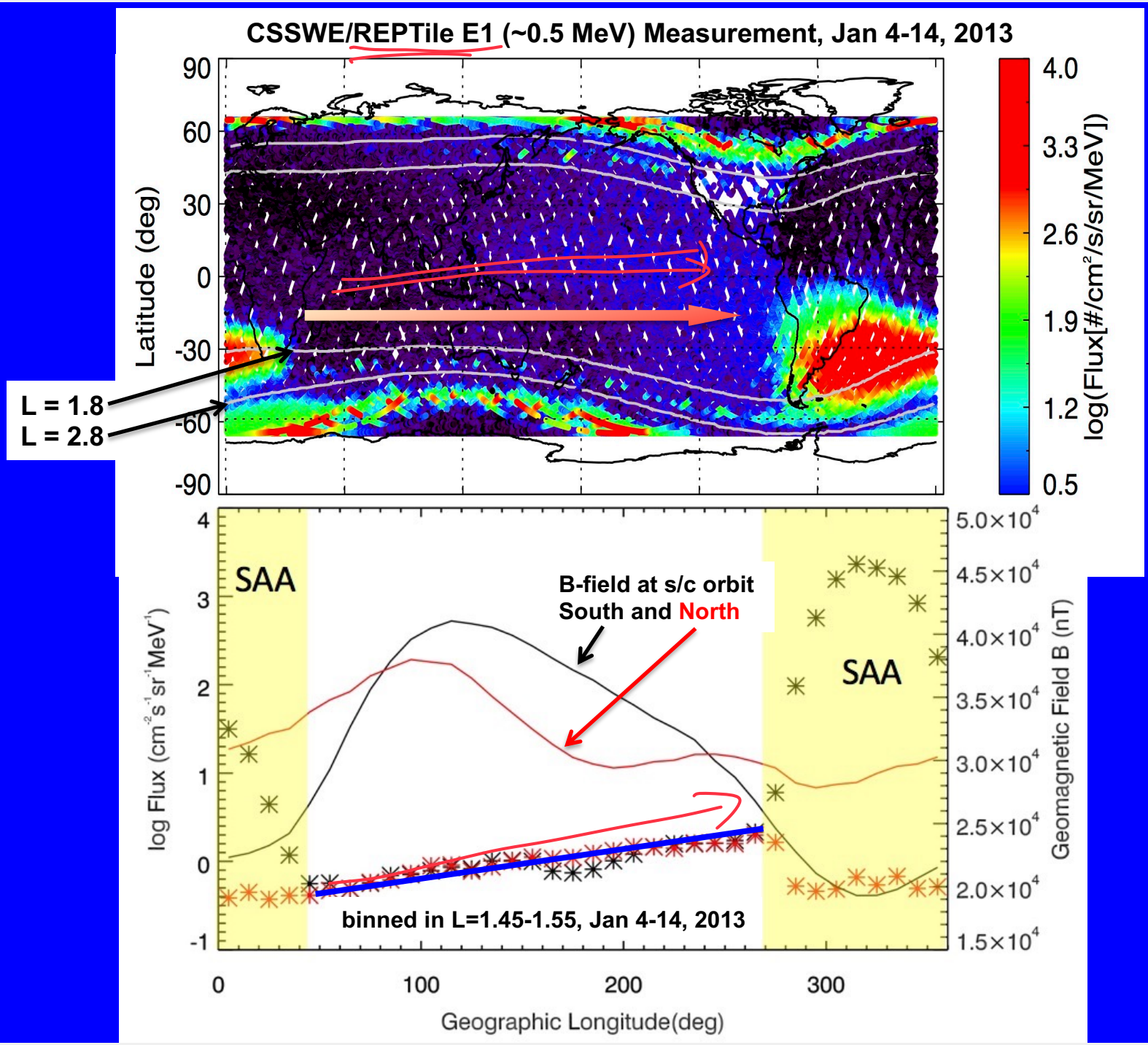
1) Li, X. (2024) Unveiling Energetic Particle Dynamics in the Near-Earth Environment From CubeSat Missions, AGU Advances, Volume 5 Issue 3, http://dx.doi.org/10.1029/2024AV001256
2) Owen, R. (2024) Tiny satellites can provide significant information about space, AGU Eos 105, https://doi.org/10.1029/2024EO240211
3) Wu J., et al. (2024) CORBES: Radiation belt survey with international small satellite constellation, *Advances in Space Research*, https://doi.org/10.1016/j.asr.2024.04.051
4) Li, X., et al. (2022) Two Generations of CubeSat Missions (CSSWE and CIRBE) to Take on the Challenges of Measuring Relativistic Electrons in the Earth's Magnetosphere, paper # SSC22-III-03, SmallSat Conference, Logan, Utah, USA, August 6-11, 2022. https://digitalcommons.usu.edu/smallsat/2022/WD-S3-2022/
5) Spence, Harlan, et al. (2022) Achievements and Lessons Learned from Successful Small Satellite Missions for Space Weather-Oriented Research, *Space Weather* [Paper #2021SW003031], https://doi.org/10.1029/2021SW003031
6) Baker, D. N., et al. (2021) The Relativistic Electron-Proton Telescope (REPT) Investigation: Design, Operational Properties, and Science highlights, Space Science Reviews (2021) 217:68, https://doi.org/10.1007/s11214-021-00838-3
7) Zhang, K., X. Li, H. Zhao, Q. Schiller, L.-Y. Khoo, Z. Xiang, R. Selesnick, M. A. Temerin, and J. A. Sauvaud (2019) Cosmic Ray Albedo Neutron Decay (CRAND) as a source of inner belt electrons: Energy spectrum study, Geophys. Res. Lett., 46, https://doi.org/10.1029/2018GL080887

8) Li, X., R. Selesnick, Q. Schiller, K. Zhang, H. Zhao, D. N. Baker, and M. Temerin (2017), Measurement of electrons from albedo neutron decay and neutron density in near-Earth space, *Nature* 552, 382-385, doi:10.1038/nature24642.

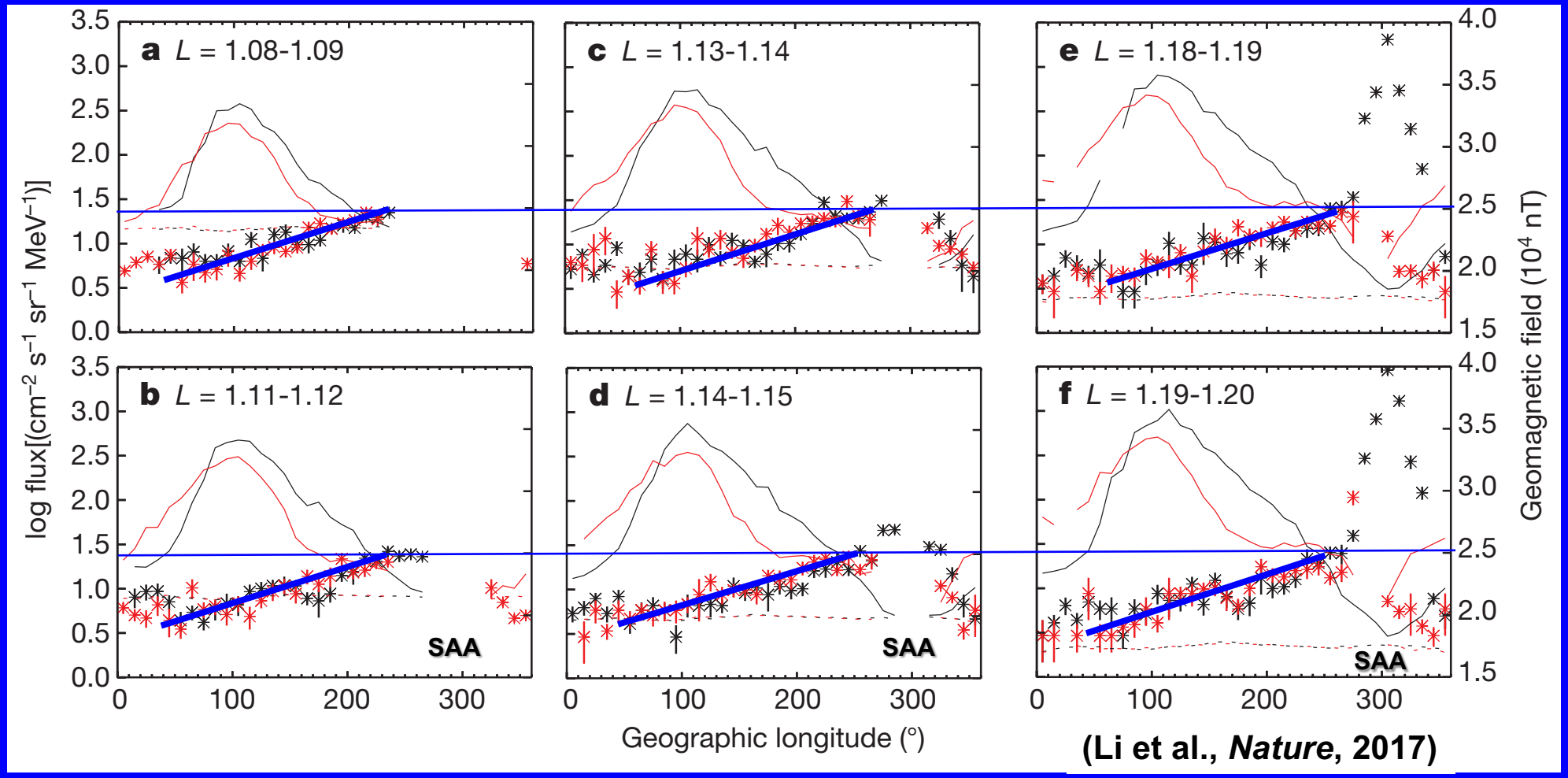
11) Q. Schiller, W. Tu, A. Ali, X. Li, H. Godinez, D. L. Turner, S. K. Morley, M. G. Henderson (2017), Simultaneous event specific estimates of transport, loss, and source rates for relativistic outer radiation belt electrons, *J. Geophys. Res. Space Physics*, 122, doi:10.1002/2016JA023093.
12) Clilverd, M. A. et al. (2017), Investigating energetic electron precipitation through combining ground-based and balloon observations, *J. Geophys. Res. Space Physics*, 122, doi:10.1002/2016JA022812.
13) Gerhardt, David T., Scott E. Palo (2016), Volume magnetization for system-level testing of magnetic materials within small satellites, *Acta Astronautica 127*, 1-12.
14) Xiang, Zheng et al. (2016), Multi-satellite simultaneous observations of magnetopause and atmospheric losses of radiation belt electrons during an intense solar wind dynamic pressure pulse, Ann. Geophys., 34, 493–509.
15) Li, X., R. S. Selesnick, D. N. Baker, A. N. Jaynes, S. G. Kanekal, Q. Schiller, L. Blum, J. F. Fennell, and J. B. Blake (2015), Upper limit on the inner radiation belt MeV electron intensity, *J. Geophys. Res. Space Physics, 120,* 1215-1228, doi:10.1002/2014JA020777.

16) Baker D. N. et al. (2014), An Impenetrable Barrier to Ultra-Relativistic Electrons in the Van Allen Radiation Belt, *Nature*, doi:10.1038/nature13956.

Experiment (CSSWE) On-Orbit Performance, *J. of Small Satellites,* Vol. 03, No. 01 (Jul 2014) pp. 265-281.
19) Schiller, Q., D. Gerhardt, L. Blum, X. Li, and S. Palo (2014), Design and Scientific Return of a Miniaturized Particle Telescope Onboard the Colorado Student Space Weather Experiment (CSSWE) CubeSat, 35th IEEE Aerospace Conference, 8.1102, doi:10.1109/AERO.2014.6836372.
20) Schiller, Q., X. Li, L. Blum, W. Tu, D. L. Turner, and J. B. Blake (2014), A non-storm time enhancement of relativistic electrons in the outer radiation belt, *Geophys. Res. Lett.,* 41, doi:10.1002/2013GL058485.
21) Blum, L. W., Q. Schiller, X. Li, R. Millan, A. Halford, and L. Woodger (2013), New conjunctive CubeSat and balloon measurements to quantify rapid energetic electron precipitation, *Geophys. Res. Lett., 40,* 1-5, doi:10.1002/2013GL058546.
22) Li, X., et al. (2013a), First results from CSSWE CubeSat: Characteristics of relativistic electrons in the near-Earth environment during the October 2012 magnetic storms, *J. Geophys. Res. Space Physics, 118,* doi:10.1002/2013JA019342.
23) Li, X., S. Palo, R. Kohnert, L. Blum, D. Gerhardt, Q. Schiller, and S. Califf (2013b), Small Mission Accomplished by Students - Big Impact on Space Weather Research, *Space Weather Journal, 11*, doi:10.1002/swe.20025, 2013.
24) Li, X., et al. (2012), Colorado Student Space Weather Experiment: Differential flux measurements of energetic particles in a highly inclined low Earth orbit, in Dynamics of the Earth's Radiation Belts and Inner Magnetosphere, Geophys. Monogr. Ser., vol. 199, edited by D. Summers et al., 385–404, AGU, Washington, D. C., doi:10.1029/2012GM001313.
25) Lauren Blum, Quintin Schiller, with advisor Xinlin Li (2012), Characterization and Testing of an Energetic Particle Telescope for a CubeSat Platform, *26th Annual AIAA/USU Conference on Small Satellites*
26) Li, X., S. Palo, and R. Kohnert (2011), Small Space Weather Research Mission Designed Fully by Students, *Space Weather Journal 9, S04006, doi:10.1029/2011SW000668*
27) Palo, S., Xinlin Li, David Gerhardt, Drew Turner, Rick Kohnert, Vaughn Hoxie and Susan Batiste (2010), Conducting Science with a CubeSat: The Colorado Student Space Weather Experiment, *24th Annual AIAA/USU Conference on Small Satellites.*
28) Schiller, Q., Abhishek Mahendrakumar, with advisor Xinlin Li (2010), REPTile: A Miniaturized Detector for a CubeSat Mission to Measure Relativistic Particles in Near-Earth Space, *24th Annual AIAA/USU Conference on Small Satellites.*
29) Gerhardt, D. T. with advisor Scott Palo (2010), Passive Magnetic Attitude Control for CubeSat Spacecraft, *24th Annual AIAA/USU Conference on Small Satellites.*


CSSWE/REPTile E1 (~0.5 MeV) Measurement, Jan 4-14, 2013
Latitude (deg)
90
60
30
0
-30
-60
-90
L = 1.8
L = 2.8
4.0
3.3
2.6
1.9
1.2
0.5
log(Flux[#/cm²/s/sr/MeV])
SAA
SAA
B-field at s/c orbit
South and North
binned in L=1.45-1.55, Jan 4-14, 2013
log Flux (cm⁻²s⁻¹sr⁻¹MeV⁻¹)
Geomagnetic Field B (nT)
5.0×10⁴
4.5×10⁴
4.0×10⁴
3.5×10⁴
3.0×10⁴
2.5×10⁴
2.0×10⁴
1.5×10⁴
4
3
2
1
0
-1
0
100
200
300
Geographic Longitude(deg)

CSSWE/REPTile: 0.5 MeV Electrons for Oct 7-10, 2012



(Li et al., *Nature*, 2017)

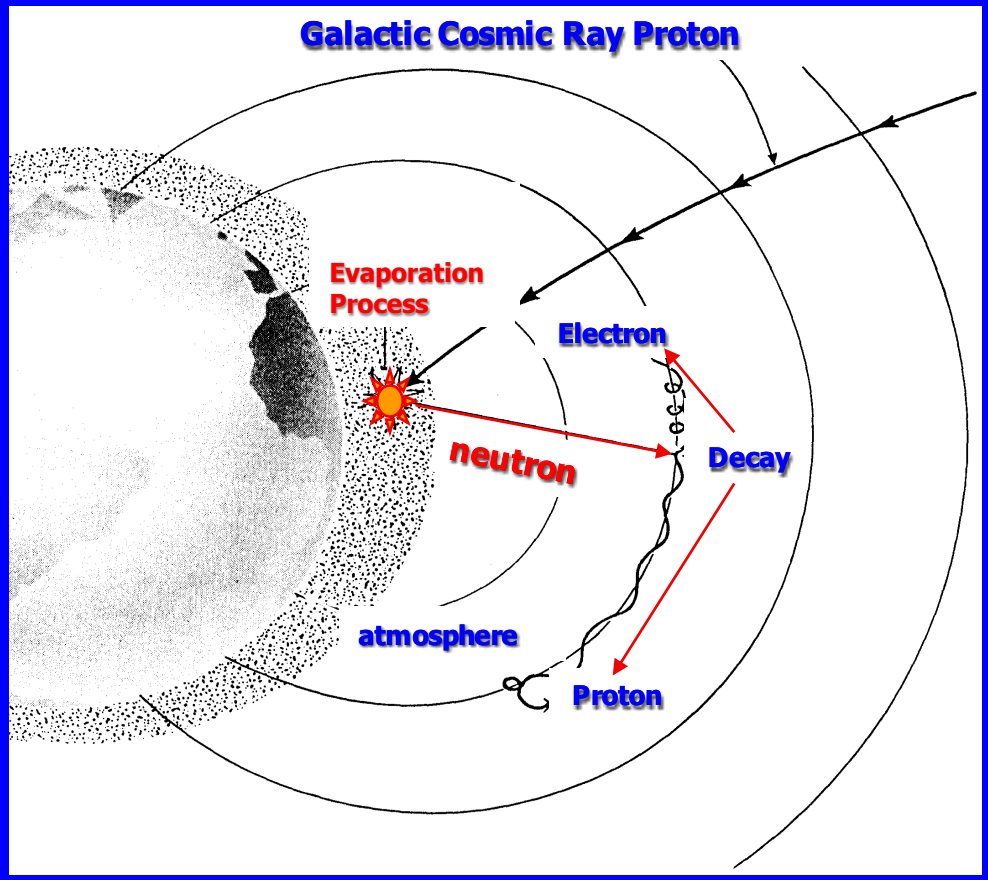
First direct detection of CRAND electrons in near-Earth space

CRAND: contributing to other regions as well, trapped and quasi-trapped

IMPLICATION: Source and Loss (pitch angle scattering) for the inner belt and slot region need to be re-visited.

Sources for Inner Belt electrons: CRAND and Inward Transport of outer belt electrons

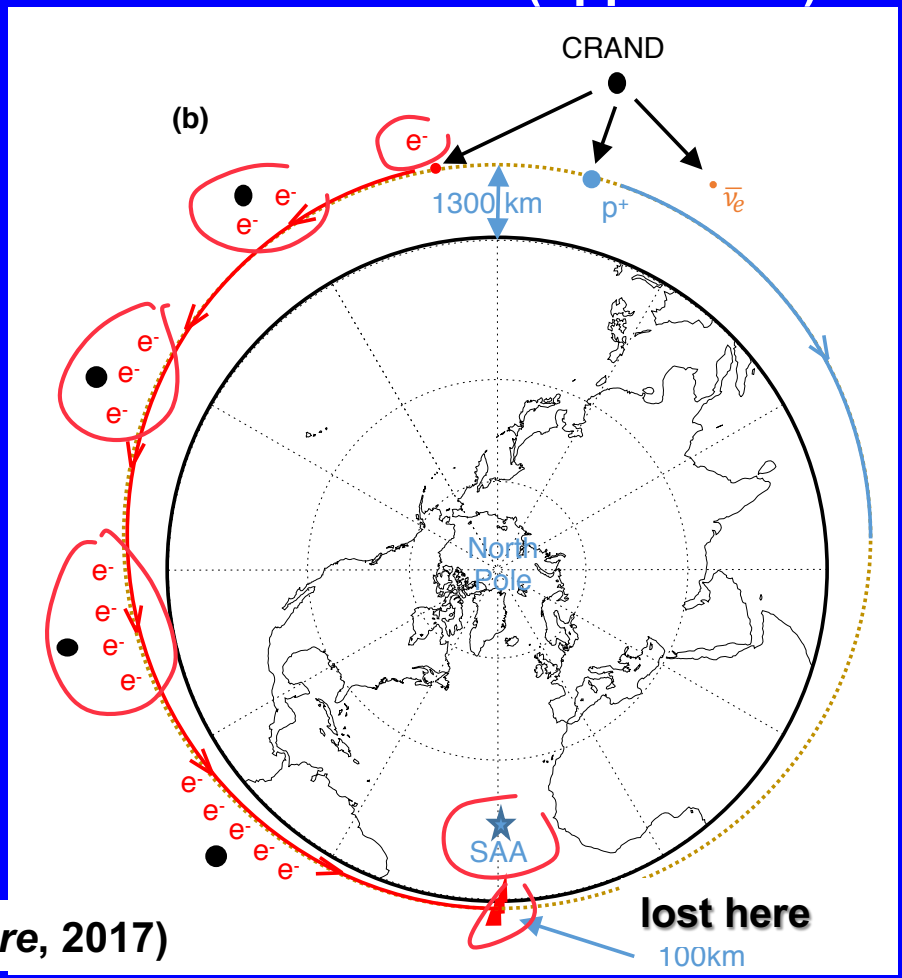
The only feasible explanation: Cosmic Ray Albedo Neutron Decay (CRAND) via evaporation process



CRAND electrons are mostly from the β-decay of thermal neutrons:

$(m_n - m_p - m_e)c^2 \approx 782$ keV

(upper limit)

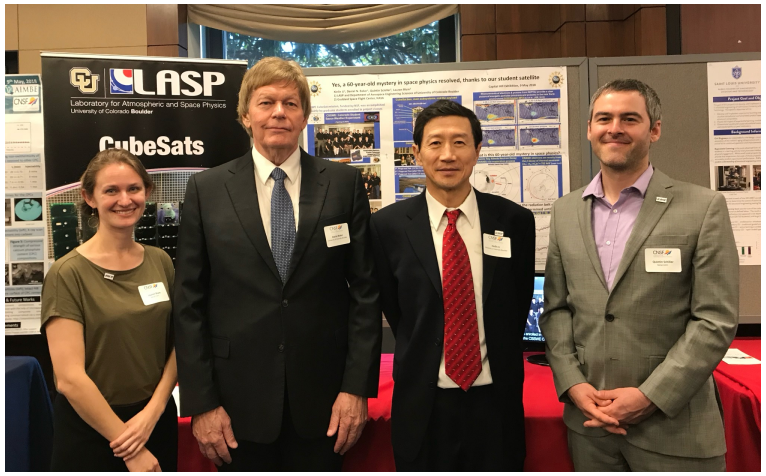


(Li et al., *Nature*, 2017)

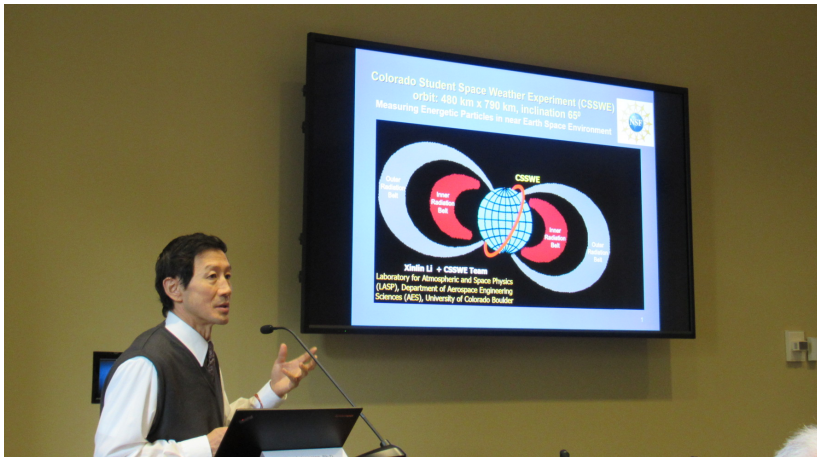
Twice Invited to Present CSSWE/REPTile Results in Capitol Hill

to advocate more funding to train the next generation of engineers and scientists

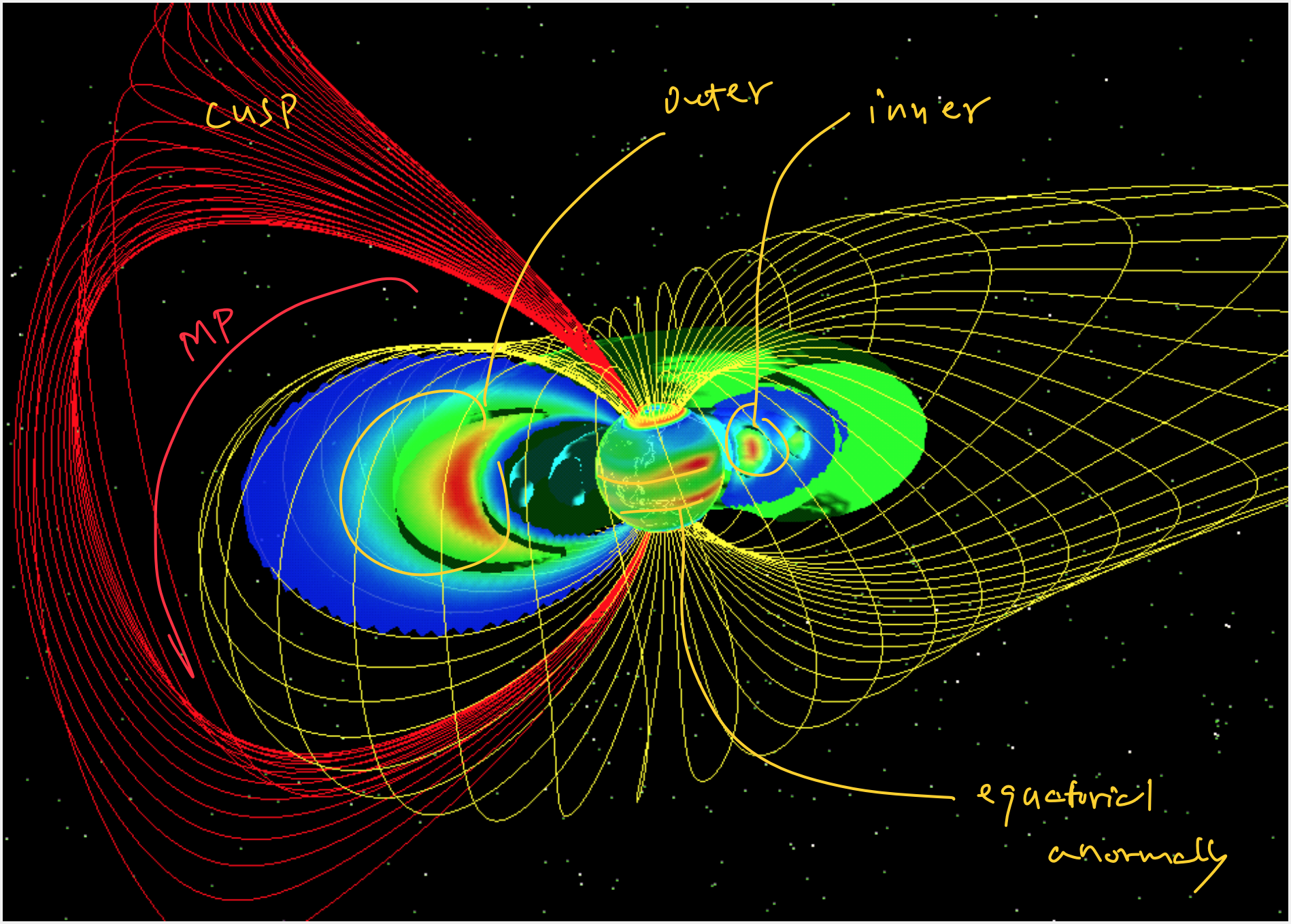
February 2018, US Geological Society



February 2019, USRA

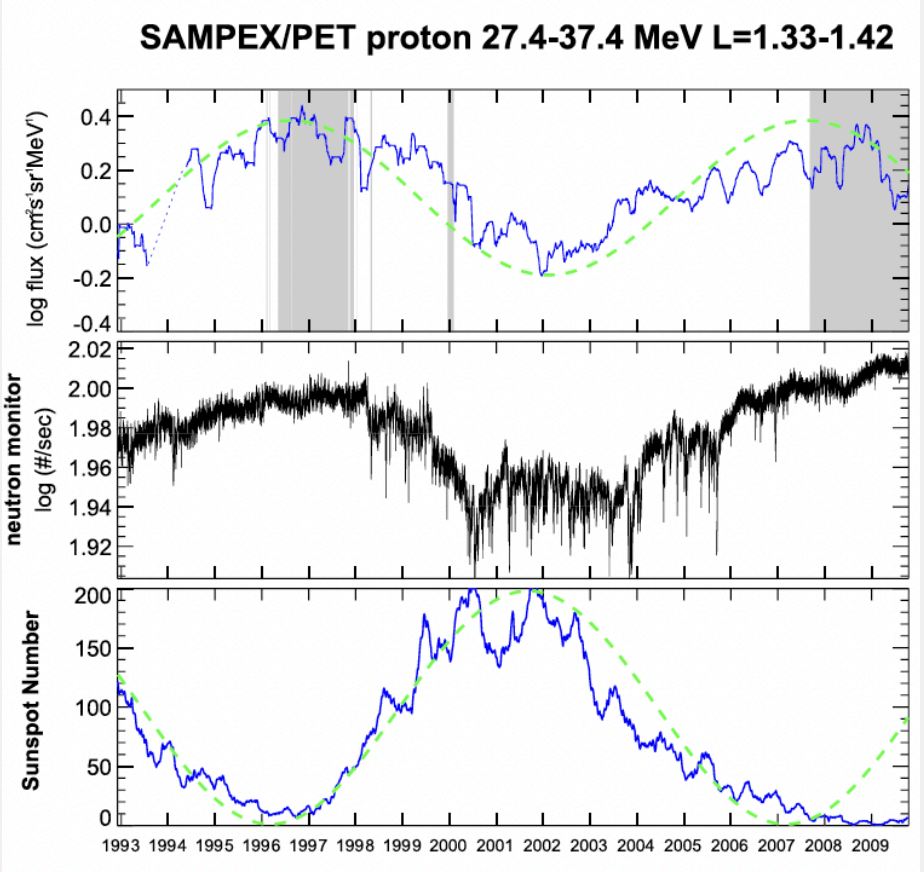


Radiation Belt Dynamics



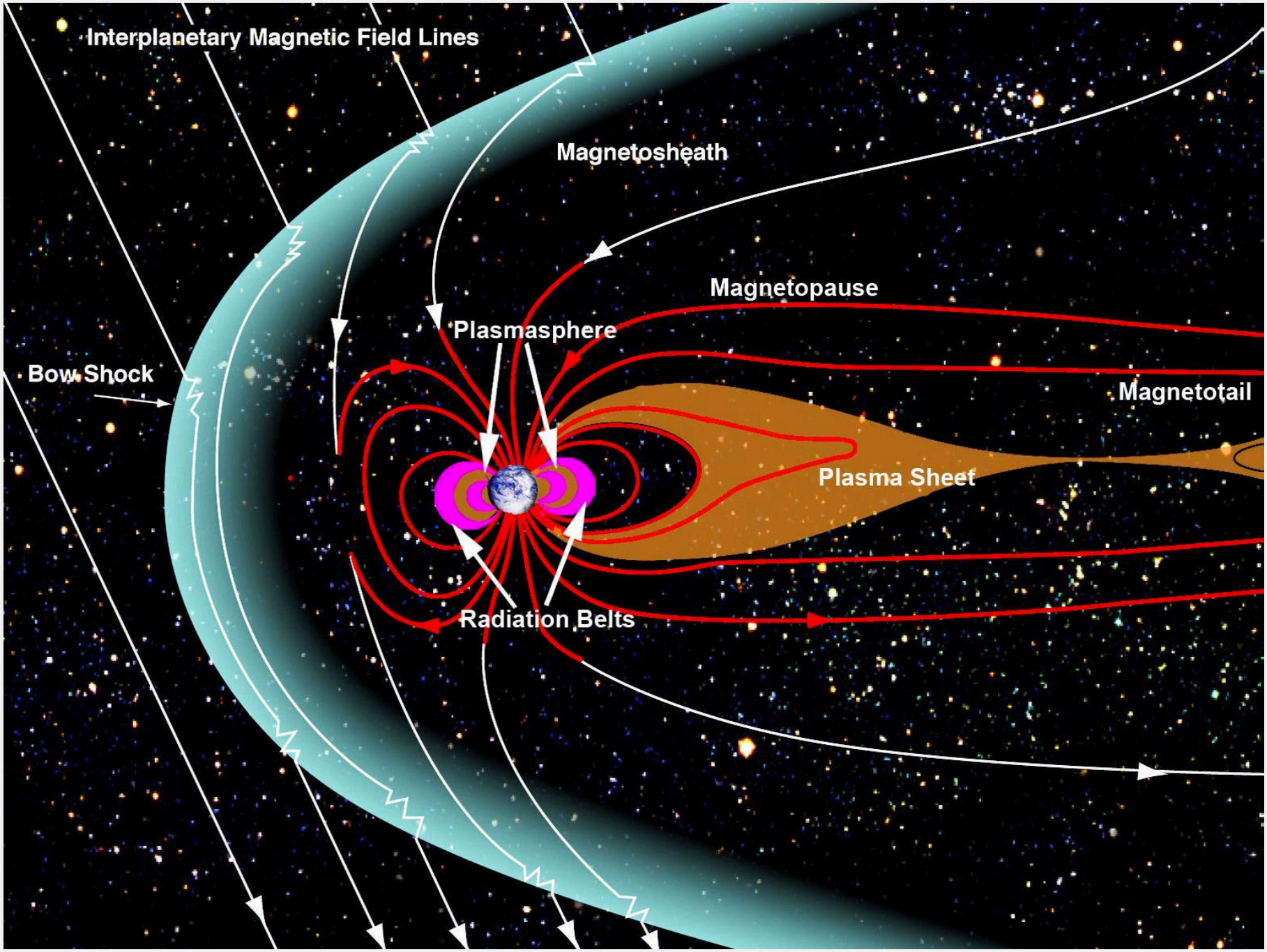
Radiation Belt Loss: Inner Belt

- Inner belt protons are lost due to charge exchange with H atoms in atmosphere
  - At first, **H^+ is fast** and **H is slow**
  - **H** gives electron to **H^+**; now **H^+ is slow** and **H is fast**
  - **H** escapes because it is not tied to B-field



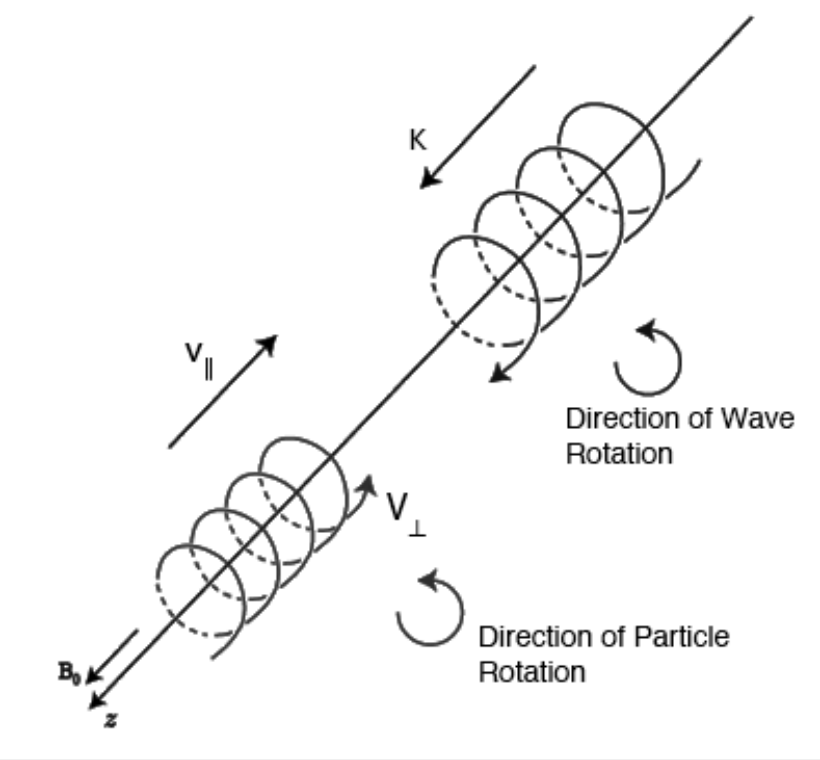
(Li et al., JGR, 2020)

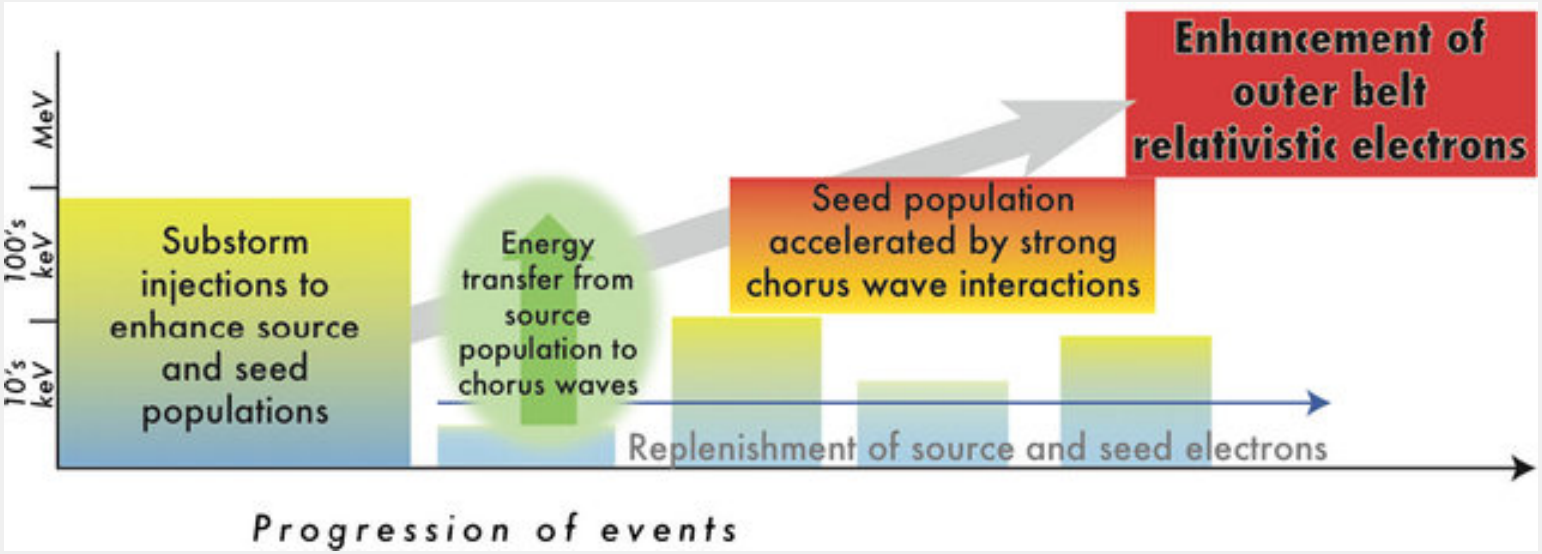
Radiation Belt Sources: Outer Belt



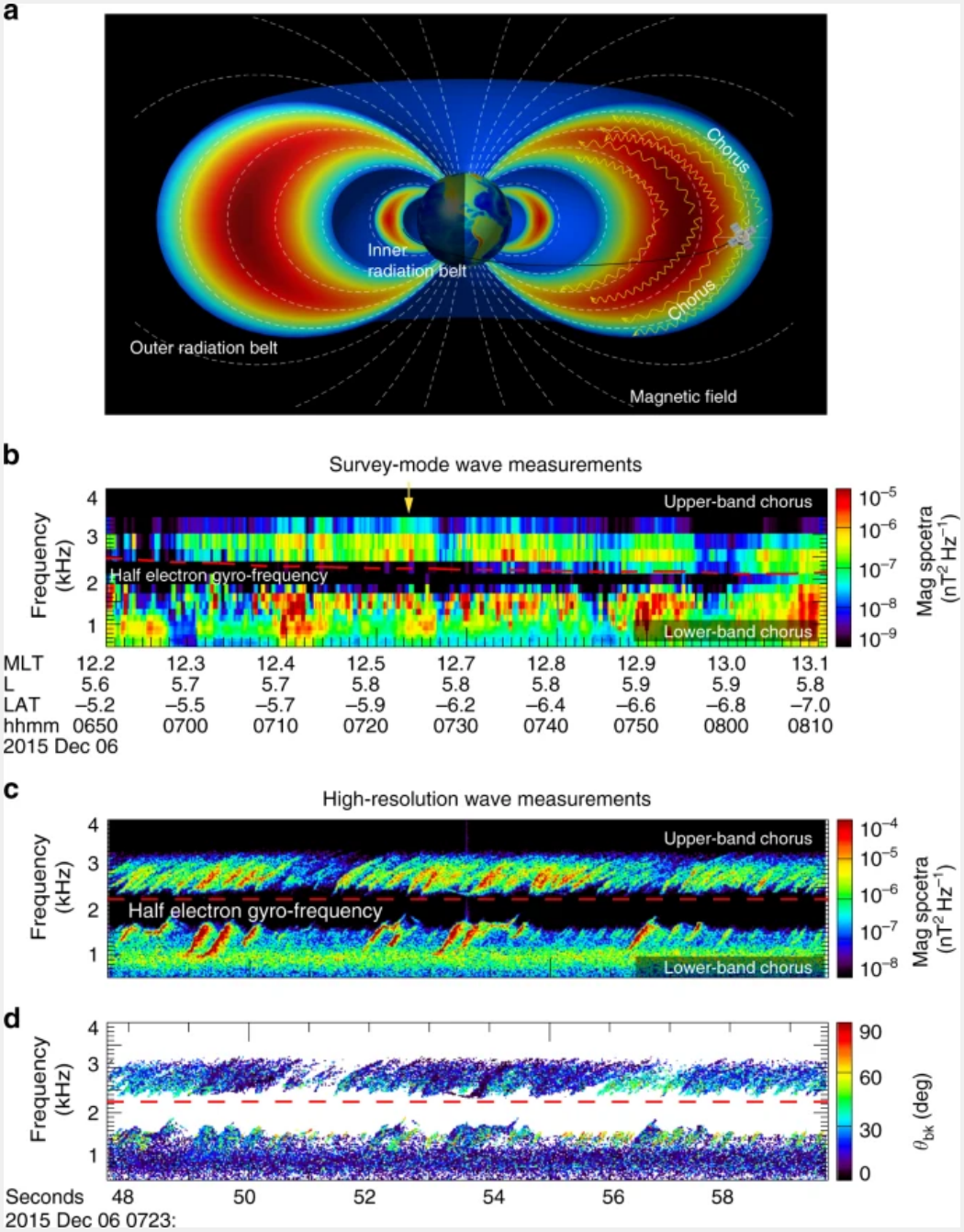
Outer Belt Energization

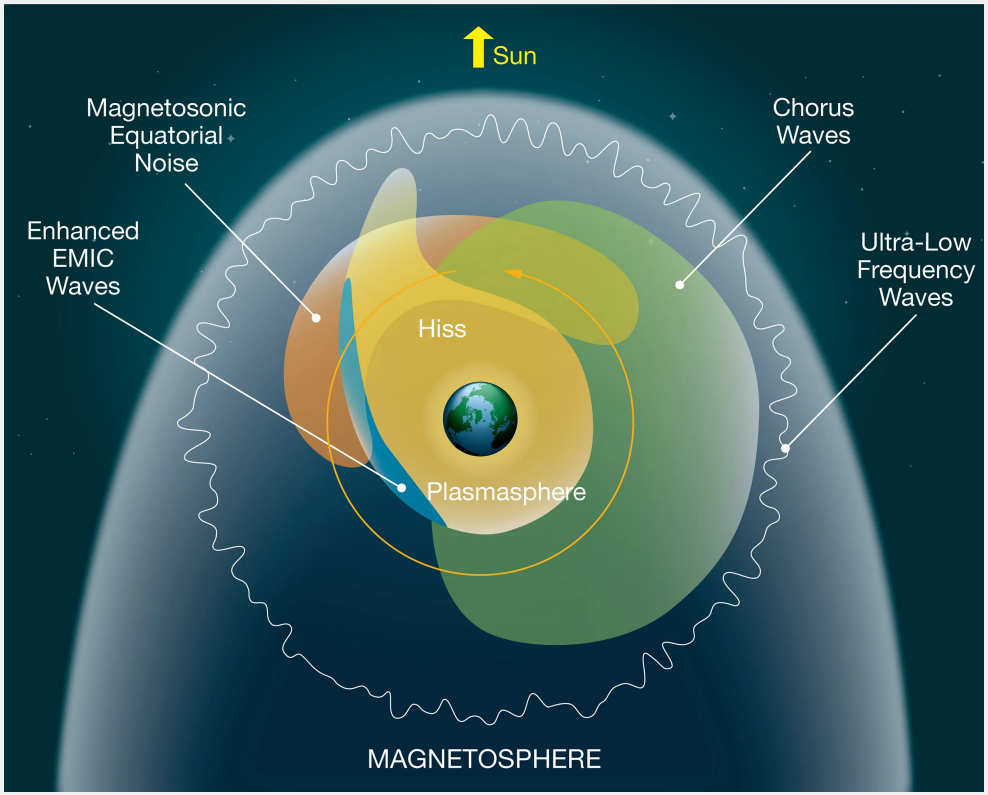
- Now we have plasma sheet electrons (10–250 keV) transported into the outer belt region (3 < L < 6):
  "seed" electrons
- Energized to MeV energies by interaction with electromagnetic waves ("chorus")
  - **"cyclotron resonance"**

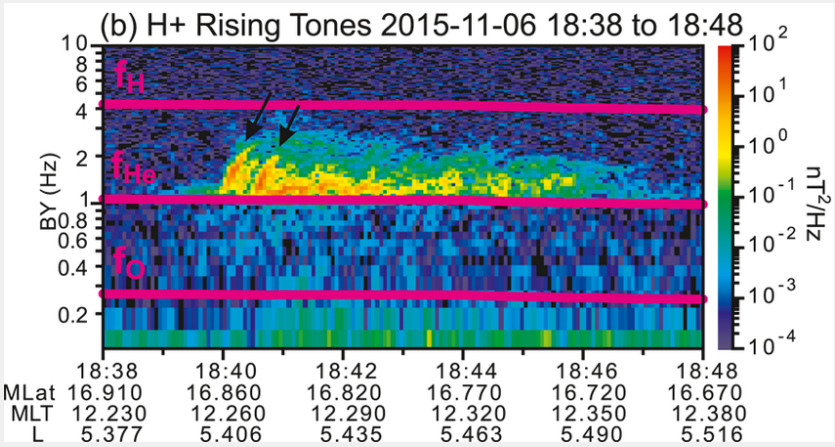




Plasma Waves and their Distributions

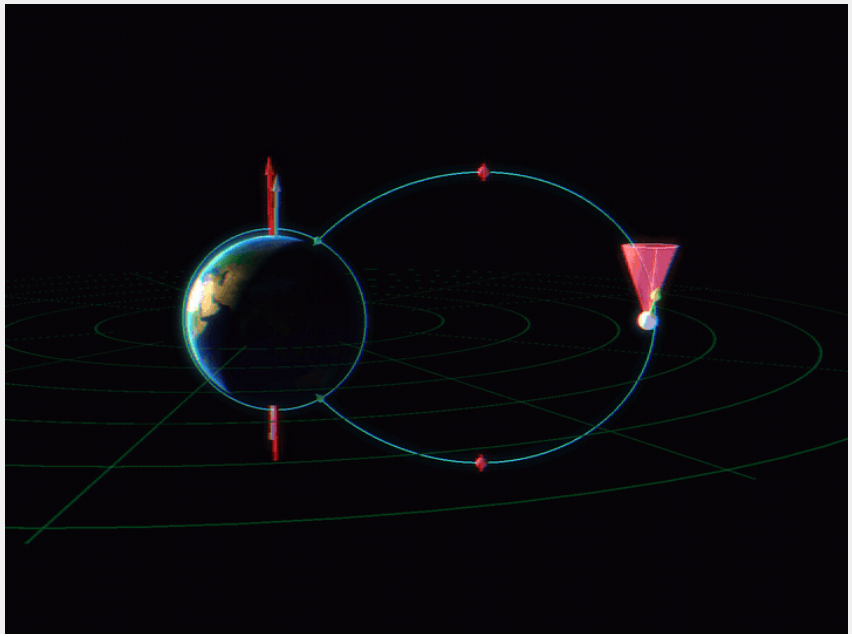


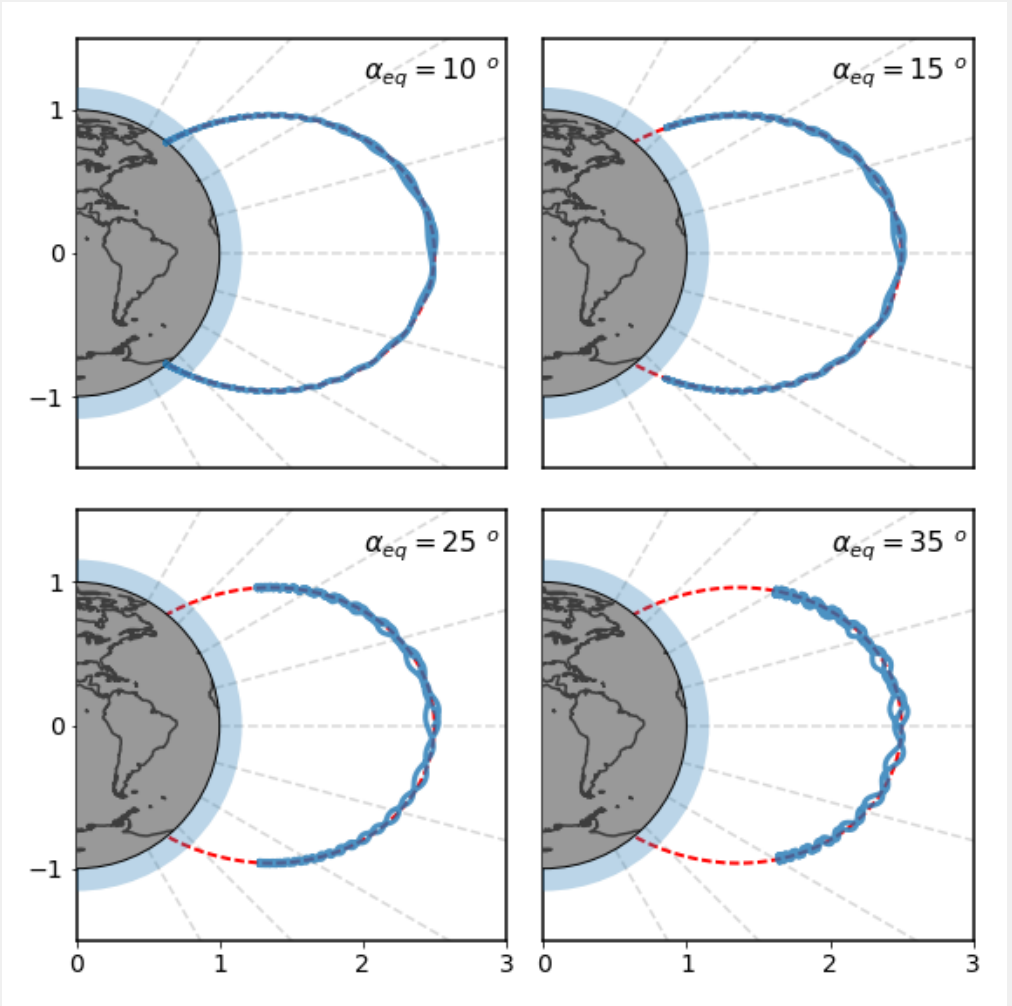




Radiation Belt Losses: Precipitation

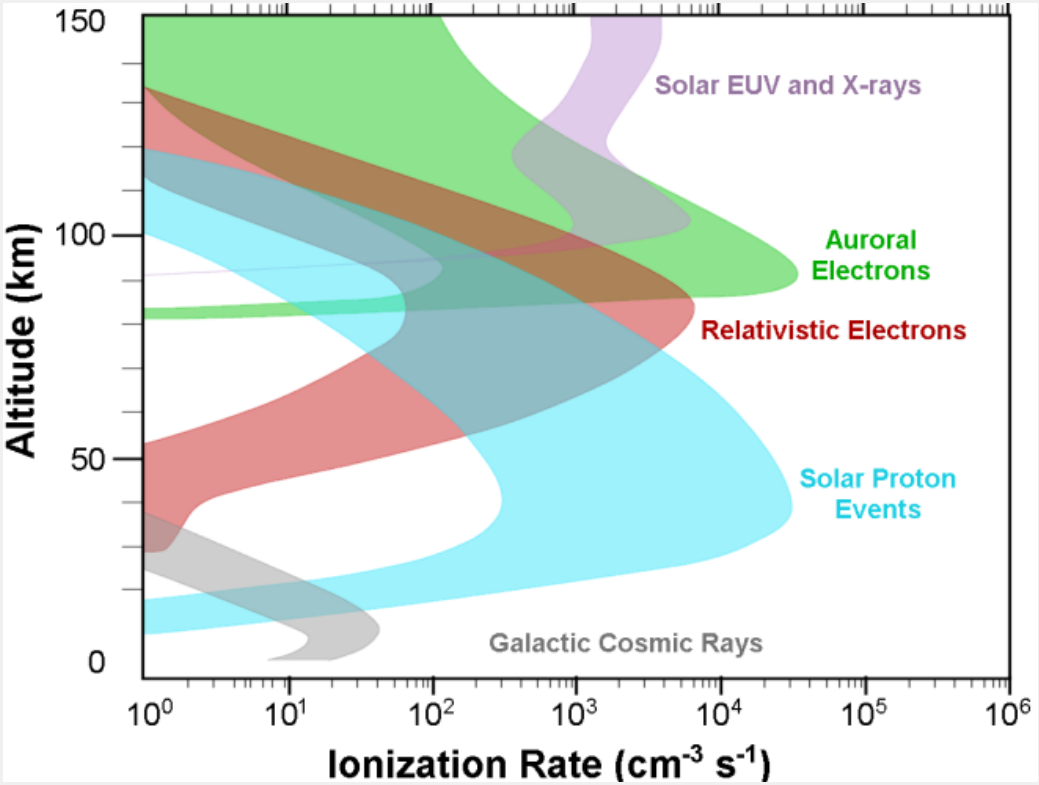
- Concept: **The Loss Cone**
  - The equatorial pitch angle α_{eq} of particles determines where they will mirror
  - α_{eq} = 90: mirroring at equator
  - α_{eq} = 45: mirroring at higher latitude
  - α_{eq} = 0: will hit the Earth!
- Loss cone angle α_{LC} – particles with lower pitch angle will reach 100 km altitude and be lost the atmosphere
- We call this **radiation belt precipitation**

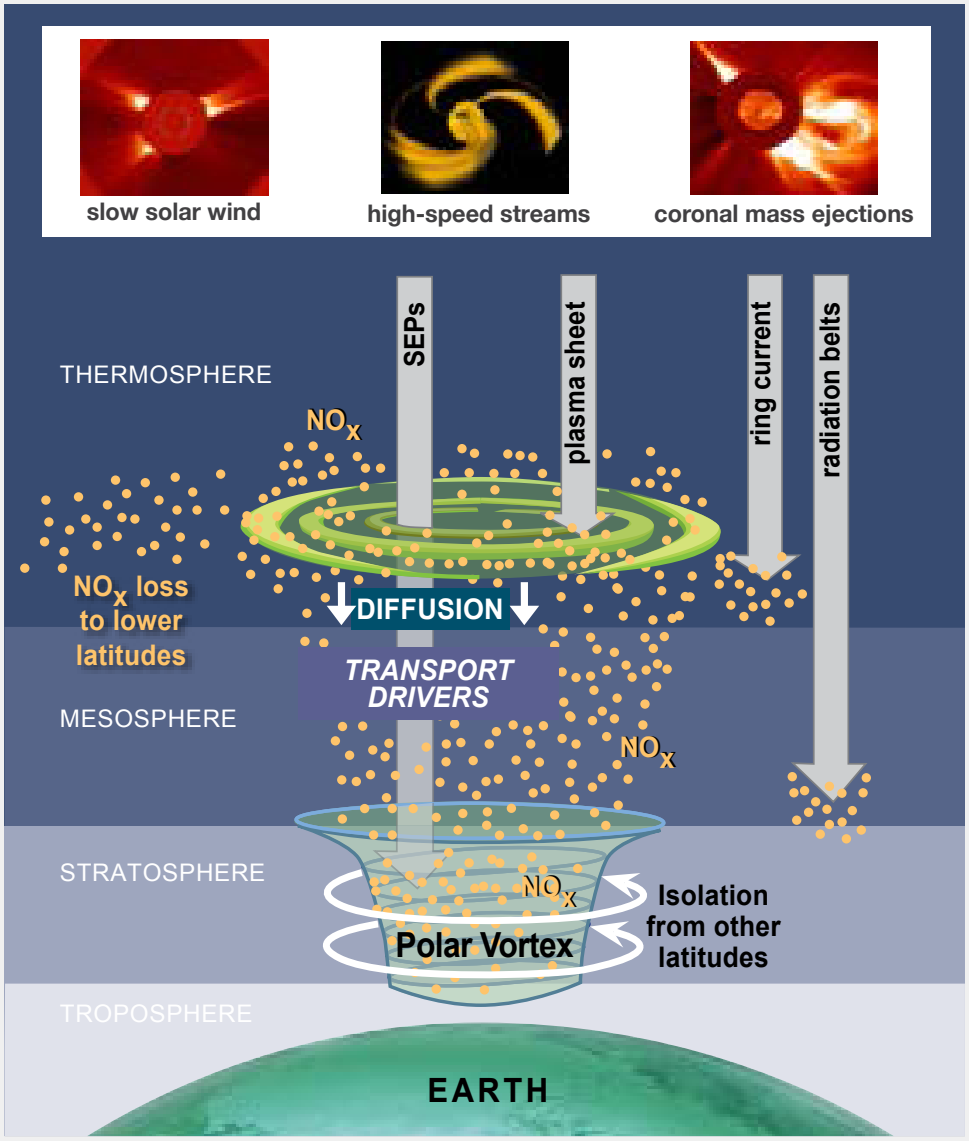




Atmospheric Effects of Precipitation

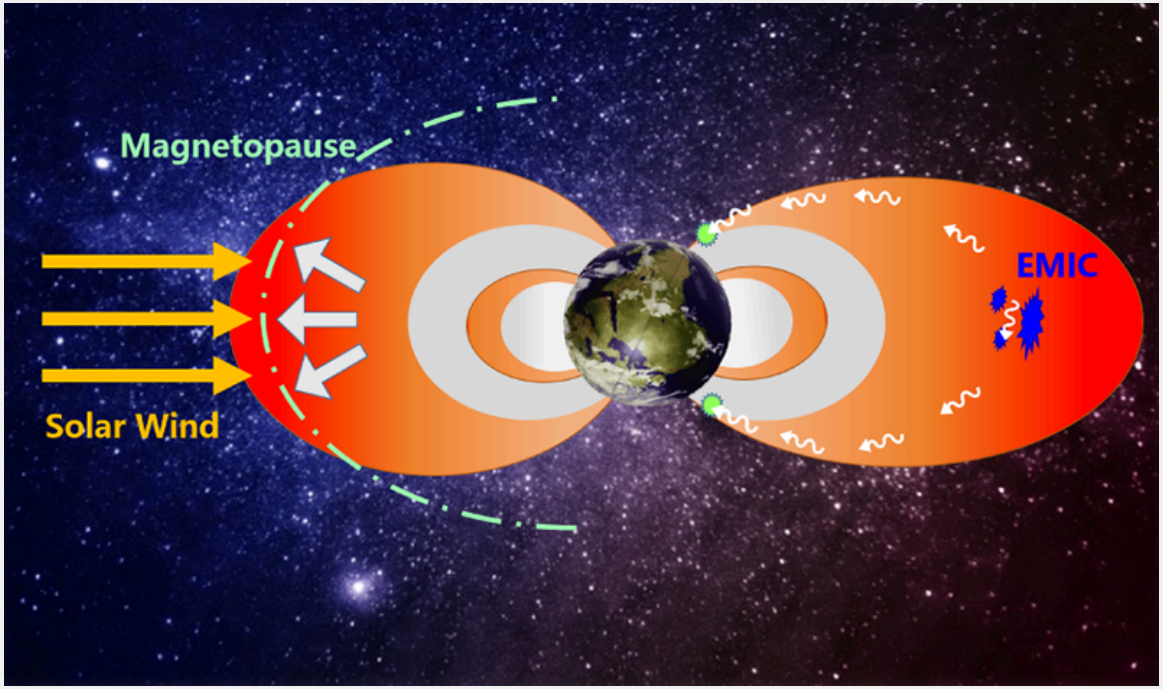
- Ionization in the upper atmosphere (temporary; recombines quickly)
- Chemical effects: NOx, HOx enhancement, ozone destruction

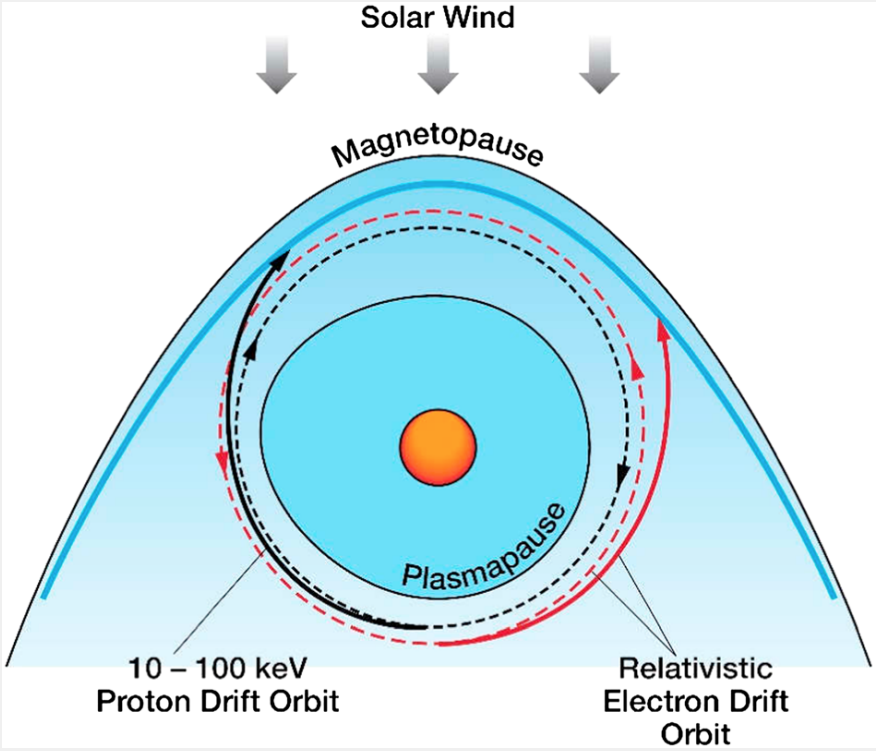




Radiation Belt Losses: Magnetopause Shadowing

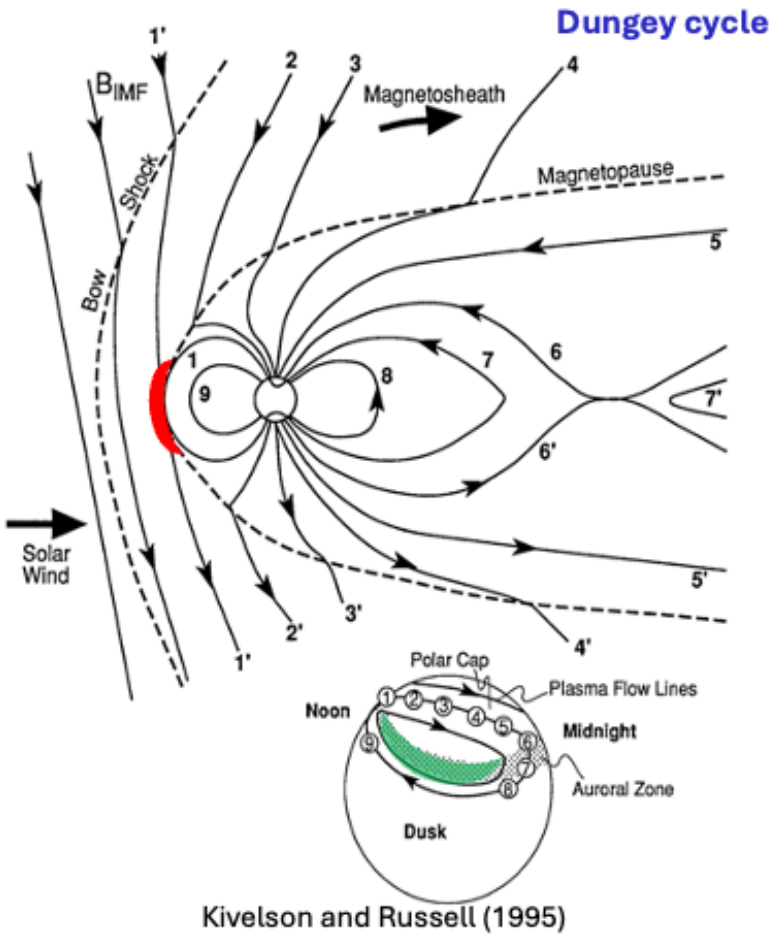
- Particles at outer edge of belt on nightside drift around to dayside; during compressed magnetosphere, outer belt may extend outside magnetopause
- Once particles reach magnetopause, they can escape on open field lines



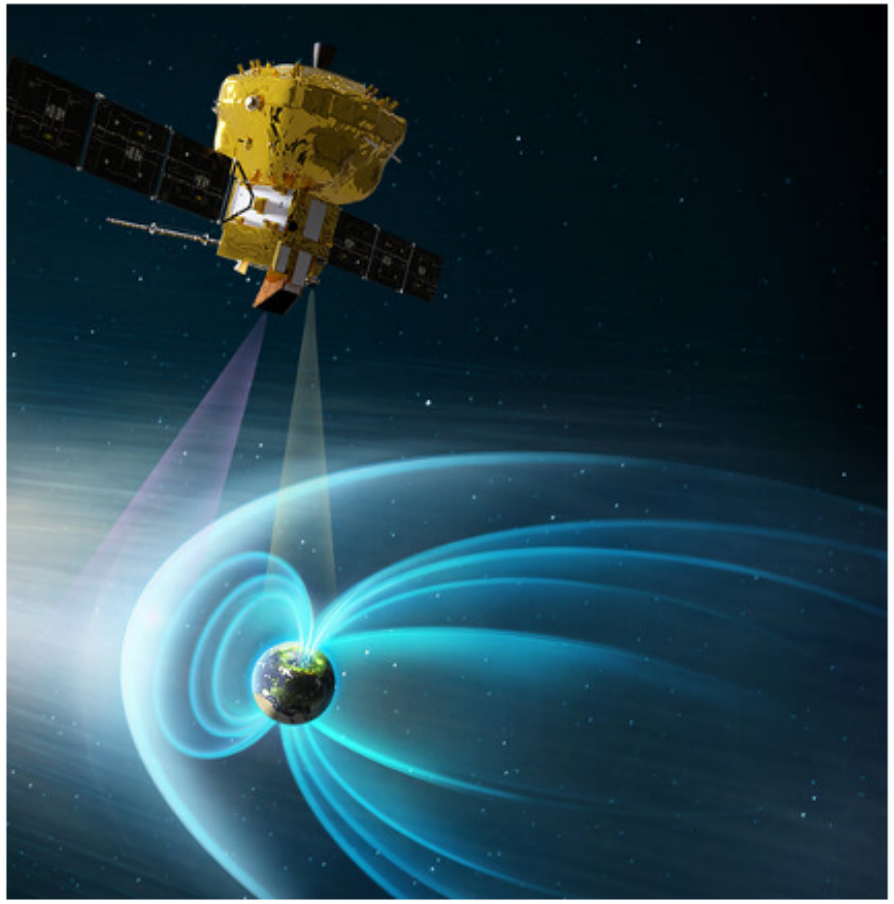


Observing Magnetopause Location

SMILE, a joint ESA–Chinese Academy of Sciences mission, was launched in May 2026 and began science operations in September 2026. Its ultraviolet imager has successfully obtained global images of the northern auroral oval, while its soft X-ray imager is being optimized to provide global observations of the dayside magnetopause and polar cusps.



Kivelson and Russell (1995)


Picture credit: ESA

University of Colorado **Boulder**

Radiation Effects

Xinlin Li